Lecture Notes on Optimization and Optimal Control

Module 4: Generalized Convex Optimization

Quanyan Zhu

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This module develops a geometric and variational framework for convex optimization that extends classical Karush–Kuhn–Tucker theory beyond standard constraint qualifications. Emphasis is placed on tangent and normal cones, feasible-direction characterizations, and saddle-point formulations of constrained convex programs. By decomposing constraints into geometric (equalities and carrier inequalities) and effective inequality components, the lectures establish saddle-point and first-order optimality conditions that remain valid even when Slater's condition fails. Restricted and complete Lagrangian formulations are introduced to clarify the role of dual variables and to recover strong duality under minimal assumptions. The theory is illustrated through examples involving degenerate constraints, affine feasible sets, entropy optimization, and convex optimal control, and shows how classical KKT conditions arise as a special case of a more general cone-based framework.

1 Generalized Saddle-Point Theorem

Let $\mathcal{P} := \{1, \ldots, p\}$, $\mathcal{Q} := \{1, \ldots, q\}$. Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{1.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is convex, $\{g_j\}_{j \in \mathcal{Q}}$ are convex inequality constraint functions, and $\{h_i\}_{i \in \mathcal{P}}$ define the equality constraints.

Throughout, we assume that the equality constraint set $\{x \in \mathbb{R}^n : h_i(x) = 0,\ i \in \mathcal{P}\}$ is a convex set. In general, even if each h_i is convex, the set $\{x : h_i(x) = 0\}$ need *not* be convex. Therefore, to ensure convexity of the feasible set, it is standard to restrict to equality constraints that are *affine*, i.e.,

$$h_i(x) = a_i^\top x - b_i, \qquad i \in \mathcal{P},$$

or, more generally, to functions whose zero level sets are convex. Affine functions are the most common and practically important case.

Under these assumptions, the feasible set of (2.60) is convex, and the problem is a convex optimization problem in the standard sense.

1.1 Feasible-Set Decomposition

All equality-type constraints are incorporated into the geometry of the feasible set. Among the inequality constraints indexed by $\mathcal{Q}$, we distinguish those that are active at every feasible point.

$$\mathcal{Q}^= := \left\{ j \in \mathcal{Q} \mid g_j(x) = 0 \ \text{ for all } x \in \mathcal{F} \right\}, \qquad \mathcal{Q}^< := \mathcal{Q} \setminus \mathcal{Q}^=. \tag{1.2}$$

We define the *equality feasible set* (base set) as

$$\mathcal{F}^= := \left\{ x \in \mathbb{R}^n \mid h_i(x) = 0,\ i \in \mathcal{P},\ \ g_j(x) = 0,\ j \in \mathcal{Q}^= \right\}. \tag{1.3}$$

The *full feasible set* is the set of points satisfying *all* constraints,

$$\mathcal{F} := \left\{ x \in \mathbb{R}^n \mid h_i(x) = 0,\ i \in \mathcal{P},\ \ g_j(x) = 0,\ j \in \mathcal{Q} \right\}. \tag{1.4}$$

Equivalently, the full feasible set can be written as

$$\mathcal{F} = \mathcal{F}^= \ \cap\ \left\{ x \in \mathbb{R}^n \mid g_j(x) = 0,\ j \in \mathcal{Q}^< \right\}. \tag{1.5}$$

1.2 General Convex Optimization Problem

Consider the convex optimization problem

$$\min_{x \in C} f(x), \tag{1.6}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is convex and differentiable, and $C \subset \mathbb{R}^n$ is a nonempty convex set.

Since both the objective function and the feasible set are convex, every local minimizer of (1.6) is also a global minimizer. Optimality can therefore be characterized using first-order information and the local geometry of the feasible set.

Theorem 1.1 (First-Order Optimality Condition: Tangent-Cone Form)**.** *A point $x^\star \in C$ is optimal for (1.6) if and only if*

$$\nabla f(x^\star)^\top d \geq 0 \quad \forall d \in T_C(x^\star), \tag{1.7}$$

where $T_C(x^\star)$ denotes the tangent (contingent) cone to C at $x^\star$.

The tangent cone $T_C(x^\star)$ captures all directions in which one can infinitesimally move from $x^\star$ while remaining feasible. Formally, it is defined as

$$T_C(x^\star) := \{d \in \mathbb{R}^n \mid \exists\, \varepsilon_k \downarrow 0,\ \exists\, d_k \to d \text{ such that } x^\star + \varepsilon_k d_k \in C \ \ \forall k\}\,. \tag{1.8}$$

Condition (1.7) states that the directional derivative of f at $x^\star$ is nonnegative along every direction that is feasible to first order. In other words, there exists no feasible descent direction at $x^\star$.

If $x^\star \in \operatorname{int}(C)$, then $T_C(x^\star) = \mathbb{R}^n$, and (1.7) reduces to

$$\nabla f(x^\star) = 0,$$

recovering the classical unconstrained first-order condition.

Equivalently, (1.7) can be expressed using the *normal cone*

$$N_C(x^\star) := \{\, v \in \mathbb{R}^n \mid v^\top d \leq 0 \ \ \forall d \in T_C(x^\star)\,\}, \tag{1.9}$$

as

$$-\nabla f(x^\star) \in N_C(x^\star). \tag{1.10}$$

This cone-based optimality condition provides the geometric foundation for Lagrangian duality, KKT conditions, and generalized saddle-point theorems in constrained convex optimization.

Proof. We prove the two directions separately.

Necessity. Assume that $x^\star \in C$ is an optimal solution of

$$\min_{x \in C} f(x),$$

where f is convex and differentiable.

Let $d \in T_C(x^\star)$ be an arbitrary tangent direction. By definition of the tangent cone, there exist sequences $\varepsilon_k \downarrow 0$ and $d_k \to d$ such that

$$x^\star + \varepsilon_k d_k \in C \quad \forall k.$$

Since $x^\star$ is optimal over C, we have

$$f(x^\star + \varepsilon_k d_k) \geq f(x^\star) \quad \forall k.$$

Dividing by $\varepsilon_k > 0$ yields

$$\frac{f(x^\star + \varepsilon_k d_k) - f(x^\star)}{\varepsilon_k} \geq 0.$$

Because f is differentiable at $x^\star$, the directional derivative exists and satisfies

$$\lim_{k \to \infty} \frac{f(x^\star + \varepsilon_k d_k) - f(x^\star)}{\varepsilon_k} = \nabla f(x^\star)^\top d.$$

Taking the limit on both sides gives

$$\nabla f(x^\star)^\top d \geq 0.$$

Since $d \in T_C(x^\star)$ was arbitrary, we conclude that

$$\nabla f(x^\star)^\top d \geq 0 \quad \forall d \in T_C(x^\star).$$

Sufficiency. Now assume that

$$\nabla f(x^\star)^\top d \geq 0 \quad \forall d \in T_C(x^\star).$$

Let $x \in C$ be arbitrary and define

$$d := x - x^\star.$$

Because C is convex, the line segment

$$x^\star + \varepsilon(x - x^\star) = (1-\varepsilon)x^\star + \varepsilon x$$

belongs to C for all $\varepsilon \in [0,1]$. Hence $d \in T_C(x^\star)$.

By the assumed condition,

$$\nabla f(x^\star)^\top (x - x^\star) \geq 0.$$

Since f is convex, it satisfies the first-order convexity inequality

$$f(x) \geq f(x^\star) + \nabla f(x^\star)^\top (x - x^\star).$$

Combining the two inequalities yields

$$f(x) \geq f(x^\star).$$

Because $x \in C$ was arbitrary, this shows that $x^\star$ is a global minimizer of f over C.

□

2 Saddle-Point Characterization without Slater's Condition

We consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{2.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is convex, $\{h_i\}_{i \in \mathcal{P}}$ are affine (or convex) equality constraint functions, and $\{g_j\}_{j \in \mathcal{Q}}$ are convex inequality constraint functions.

Feasible-Set Decomposition

All equality constraints are incorporated into the geometry of the feasible set. Among the inequality constraints indexed by $\mathcal{Q}$, we distinguish those that are active at every feasible point.

Define the index sets

$$\mathcal{Q}^{=} := \{\, j \in \mathcal{Q} \mid g_j(x) = 0 \ \text{ for all } x \in \mathcal{F} \,\}, \qquad \mathcal{Q}^{<} := \mathcal{Q} \setminus \mathcal{Q}^{=}. \tag{2.2}$$

Equivalently, for each $j \in \mathcal{Q}^{<}$ there exists at least one feasible point $x' \in \mathcal{F}$ such that $g_j(x') < 0$.

The *base feasible set* is defined by all equality constraints together with the inequality constraints that behave as equalities:

$$\mathcal{F}^{=} := \left\{ x \in \mathbb{R}^n \;\middle|\; h_i(x) = 0,\ i \in \mathcal{P},\ \ g_j(x) = 0,\ j \in \mathcal{Q}^{=} \right\}. \tag{2.3}$$

The set $\mathcal{F}^{=}$ is convex (typically affine).

The *full feasible set* of (2.60) is

$$\mathcal{F} = \mathcal{F}^{=} \ \cap\ \left\{ x \in \mathbb{R}^n \mid g_j(x) \le 0,\ j \in \mathcal{Q}^{<} \right\}. \tag{2.4}$$

The constraints indexed by $\mathcal{Q}^{<}$ are referred to as the *effective inequality constraints*. No constraint qualification, including Slater's condition, is assumed.

We define the *restricted Lagrangian* by introducing multipliers only for the effective inequality constraints:

$$\mathcal{L}^c(x, \mu) := f(x) + \sum_{j \in \mathcal{Q}^{<}} \mu_j g_j(x), \quad \mu \in \mathbb{R}_+^{|\mathcal{Q}^{<}|}, \tag{2.5}$$

with the understanding that $x \in \mathcal{F}^{=}$.

Theorem 2.1 (Saddle–Point Characterization without Slater's Condition)**.** *Consider the convex optimization problem (2.60). Let $\mathcal{F}^{=}$ denote the base feasible set and let $\mathcal{Q}^{<}$ be the index set of effective inequality constraints, with $|\mathcal{Q}^{<}| = c$. Assume that $\mathcal{F}^{=}$ is nonempty.*

A point $x^\star \in \mathcal{F}^{=}$ is an optimal solution of (2.60) if and only if there exists a multiplier vector $\mu^\star \in \mathbb{R}_+^c$ such that $(x^\star, \mu^\star)$ is a saddle point of the restricted Lagrangian $\mathcal{L}^c$ over $\mathcal{F}^{=} \times \mathbb{R}_+^c$, that is,

$$\mathcal{L}^c(x^\star, \mu) \ \le\ \mathcal{L}^c(x^\star, \mu^\star) \ \le\ \mathcal{L}^c(x, \mu^\star), \quad \forall \mu \in \mathbb{R}_+^c,\ \forall x \in \mathcal{F}^{=}. \tag{2.6}$$

Proof. We prove necessity and sufficiency separately.

Necessity. Assume that $x^\star \in \mathcal{F}^=$ is an optimal solution of (2.60). Without loss of generality, relabel the effective inequality constraints so that

$$\mathcal{Q}^< = \{1, 2, \ldots, c\}.$$

Define the sets $C_1, C_2 \subset \mathbb{R}^{c+1}$ by

$$C_1 := \left\{ y \in \mathbb{R}^{c+1} \mid y \geq \begin{pmatrix} f(x) \\ g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix} \text{ for some } x \in \mathcal{F}^= \right\}, \tag{2.7}$$

$$C_2 := \left\{ y \in \mathbb{R}^{c+1} \mid y < \begin{pmatrix} f(x^\star) \\ 0 \\ \vdots \\ 0 \end{pmatrix} \right\}, \tag{2.8}$$

where inequalities are understood componentwise.

Since $\mathcal{F}^=$ is convex and the functions $f, g_1, \ldots, g_c$ are convex, the set C_1 is convex. The set C_2 is convex and open. Moreover, $C_1 \cap C_2 = \varnothing$. Indeed, if $y \in C_1 \cap C_2$, then there exists $x \in \mathcal{F}^=$ such that

$$f(x) < f(x^\star), \qquad g_j(x) < 0, \quad j = 1, \ldots, c,$$

which implies that x is feasible for (2.60) and yields a strictly smaller objective value, contradicting the optimality of $x^\star$.

By the Hyperplane Separation Theorem, there exist a nonzero vector

$$a = (a_0, a_1, \ldots, a_c) \in \mathbb{R}^{c+1} \quad \text{and} \quad \alpha \in \mathbb{R}$$

such that

$$\langle a, y^2 \rangle \leq \alpha \leq \langle a, y^1 \rangle \quad \forall y^1 \in C_1,\ \forall y^2 \in C_2. \tag{2.9}$$

Since C_2 is unbounded below in every coordinate, it follows that

$$a_i \geq 0, \quad i = 0, 1, \ldots, c.$$

Choose

$$y^1 = \begin{pmatrix} f(x) \\ g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix}, \quad x \in \mathcal{F}^=, \qquad y^2 = \begin{pmatrix} f(x^\star) \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Then (2.9) yields

$$a_0 f(x^\star) \;\leq\; a_0 f(x) + \sum_{j=1}^{c} a_j g_j(x), \quad \forall x \in \mathcal{F}^=. \tag{2.10}$$

We claim that $a_0 > 0$. Otherwise, if $a_0 = 0$, then at least one $a_j > 0$, which would imply

$$\sum_{j=1}^{c} a_j g_j(x) \geq 0 \quad \forall x \in \mathcal{F}^=,$$

contradicting the definition of $\mathcal{Q}^<$, since for each $j \in \mathcal{Q}^<$ there exists $x' \in \mathcal{F}^=$ such that $g_j(x') < 0$.

Hence $a_0 > 0$. Define the multipliers

$$\mu_j^\star := \frac{a_j}{a_0}, \quad j = 1, \ldots, c.$$

Dividing (2.10) by a_0 yields

$$f(x^\star) \;\leq\; f(x) + \sum_{j=1}^{c} \mu_j^\star g_j(x), \quad \forall x \in \mathcal{F}^=, \tag{2.11}$$

that is,

$$\mathcal{L}^c(x^\star, \mu^\star) \;\leq\; \mathcal{L}^c(x, \mu^\star), \quad \forall x \in \mathcal{F}^=.$$

Moreover, since $x^\star$ is feasible and $\mu \geq 0$,

$$\mathcal{L}^c(x^\star, \mu) = f(x^\star) + \sum_{j=1}^{c} \mu_j g_j(x^\star) \leq f(x^\star) = \mathcal{L}^c(x^\star, \mu^\star), \quad \forall \mu \geq 0.$$

Thus $(x^\star, \mu^\star)$ satisfies the saddle-point inequalities.

Sufficiency. Conversely, assume that there exists $\mu^\star \in \mathbb{R}^c_+$ such that

$$\mathcal{L}^c(x^\star, \mu) \leq \mathcal{L}^c(x^\star, \mu^\star) \leq \mathcal{L}^c(x, \mu^\star), \quad \forall \mu \geq 0, \; \forall x \in \mathcal{F}^=.$$

First, we show that $x^\star$ is feasible for the inequality constraints. If $g_k(x^\star) > 0$ for some $k \in \mathcal{Q}^<$, choose $\mu_k \to \infty$ and $\mu_j = 0$ for $j \neq k$, which violates the left inequality. Hence $g_j(x^\star) \leq 0$ for all j.

Next, setting $\mu = 0$ in the right inequality yields

$$f(x^\star) \leq f(x) + \sum_{j=1}^{c} \mu_j^\star g_j(x), \quad \forall x \in \mathcal{F}^=.$$

For any feasible x, $g_j(x) \leq 0$ for all j and $\mu_j^\star \geq 0$, implying $f(x^\star) \leq f(x)$. Therefore $x^\star$ is optimal.

□

Remark 2.2 (Saddle Point and Complementary Slackness). A pair $(x^\star, \mu^\star)$ satisfying (2.37) is called a *saddle point* of the restricted Lagrangian $\mathcal{L}^c$ over $\mathcal{F}^= \times \mathbb{R}^c_+$.

(i) From the saddle-point inequalities and feasibility of $x^\star$, it follows that

$$\mu_j^\star \, g_j(x^\star) = 0, \qquad j \in \mathcal{Q}^<. \tag{2.12}$$

These are the *complementary slackness conditions*. They express the fact that only active inequality constraints influence the optimality conditions: if $g_j(x^\star) < 0$ for some $j \in \mathcal{Q}^<$, then necessarily $\mu_j^\star = 0$.

(ii) The restricted Lagrangian $\mathcal{L}^c$ can equivalently be replaced by the classical Lagrangian

$$\mathcal{L}(x, \mu) = f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x),$$

without affecting the saddle-point characterization, provided that the multipliers satisfy $\mu_j = 0$ for all $j \in \mathcal{Q}^=$. Indeed, by definition of $\mathcal{Q}^=$, each constraint g_j with $j \in \mathcal{Q}^=$ satisfies

$$g_j(x) = 0 \qquad \text{for all } x \in \mathcal{F}^=.$$

Consequently, the corresponding Lagrange terms vanish identically on the domain $\mathcal{F}^=$, and the classical and restricted Lagrangians coincide on $\mathcal{F}^= \times \mathbb{R}^c_+$ once $\mu_j = 0$ is enforced for $j \in \mathcal{Q}^=$. These constraints therefore play a purely geometric role in defining the feasible set and do not contribute additional dual degrees of freedom.

(iii) The separating set used in the proof of Theorem 2.1 depends only on the feasible set $\mathcal{F}^=$

and not on the particular choice of an optimal solution. Consequently, the set of multipliers $\mu^\star$ satisfying the saddle-point condition is the same for all optimal solutions of (2.60).

Corollary 2.3 (Saddle-Point Characterization under Slater's Condition)**.** *Consider the convex optimization problem (2.60). Assume that the inequality constraints satisfy Slater's condition, i.e., there exists $\bar{x} \in \mathbb{R}^n$ such that*

$$h_i(\bar{x}) = 0, \quad i \in \mathcal{P}, \qquad g_j(\bar{x}) < 0, \quad j \in \mathcal{Q}.$$

Then $\mathcal{Q}^= = \varnothing$, $\mathcal{F}^= = \mathbb{R}^n$, and the restricted Lagrangian coincides with the classical Lagrangian. A point $x^\star \in \mathbb{R}^n$ is optimal if and only if there exists $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}|}$ such that

$$\mathcal{L}(x^\star, \mu) \;\leq\; \mathcal{L}(x^\star, \mu^\star) \;\leq\; \mathcal{L}(x, \mu^\star), \quad \forall \mu \in \mathbb{R}_+^{|\mathcal{Q}|},\ \forall x \in \mathbb{R}^n.$$

In this case, the saddle-point characterization reduces to the classical KKT conditions.

2.1 First-Order Optimality without Slater's Condition

We now derive a first-order characterization of optimality that does not rely on any constraint qualification, including Slater's condition. The result is a direct geometric consequence of the saddle-point characterization established earlier and expresses optimality in terms of normal and tangent cones.

Recall that $\mathcal{F}^=$ denotes the base feasible set induced by all equality-type constraints, and that $\mathcal{Q}^<$ indexes the effective inequality constraints.

Theorem 2.4 (First-Order Optimality without Slater)**.** *Consider the convex optimization problem (2.60). Assume that all functions are differentiable at a feasible point $x^\star \in \mathcal{F}^=$. Then, $x^\star$ is an optimal solution of (2.60) if and only if there exists a multiplier vector $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that*

$$-\nabla f(x^\star) - \sum_{j \in \mathcal{Q}^<} \mu_j^\star \nabla g_j(x^\star) \;\in\; N_{\mathcal{F}^=}(x^\star), \tag{2.13}$$

where $N_{\mathcal{F}^=}(x^\star)$ denotes the normal cone to $\mathcal{F}^=$ at $x^\star$.

Using the polarity relation between normal and tangent cones, condition (2.13) is equivalent to

$$\left[\nabla f(x^\star) + \sum_{j \in \mathcal{Q}^<} \mu_j^\star \nabla g_j(x^\star)\right]^\top d \geq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^\star), \tag{2.14}$$

where $T_{\mathcal{F}^=}(x^\star)$ is the tangent cone to $\mathcal{F}^=$ at $x^\star$.

Proof. We prove necessity and sufficiency separately.

Necessity. Assume that $x^\star$ is an optimal solution of (2.60). By the saddle-point characterization without Slater's condition (Theorem 2.1), there exists $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that

$$\mathcal{L}^c(x^\star, \mu) \leq \mathcal{L}^c(x^\star, \mu^\star) \leq \mathcal{L}^c(x, \mu^\star) \quad \forall \mu \geq 0,\ \forall x \in \mathcal{F}^=.$$

In particular, the right-hand inequality implies that $x^\star$ minimizes the convex function

$$x \mapsto \mathcal{L}^c(x, \mu^\star) = f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j^\star g_j(x)$$

over the convex set $\mathcal{F}^=$.

Since $\mathcal{L}^c(\cdot, \mu^\star)$ is differentiable at $x^\star$, the first-order optimality condition for convex minimization over a convex set yields

$$\nabla_x \mathcal{L}^c(x^\star, \mu^\star)^\top d \geq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^\star).$$

Expanding the gradient gives

$$\left[\nabla f(x^\star) + \sum_{j \in \mathcal{Q}^<} \mu_j^\star \nabla g_j(x^\star)\right]^\top d \geq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^\star),$$

which is precisely (2.14). By definition of the normal cone, this is equivalent to (2.13).

Sufficiency. Conversely, suppose there exists $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that (2.13) holds. Equivalently, (2.14) is satisfied.

Then, for every $x \in \mathcal{F}^=$, convexity of $\mathcal{L}^c(\cdot, \mu^\star)$ implies

$$\mathcal{L}^c(x, \mu^\star) \geq \mathcal{L}^c(x^\star, \mu^\star) + \nabla_x \mathcal{L}^c(x^\star, \mu^\star)^\top (x - x^\star).$$

Since $x - x^\star \in T_{\mathcal{F}^=}(x^\star)$, the second term on the right-hand side is nonnegative by (2.14). Hence

$$\mathcal{L}^c(x, \mu^\star) \geq \mathcal{L}^c(x^\star, \mu^\star) \quad \forall x \in \mathcal{F}^=.$$

Together with feasibility of $x^\star$ and nonnegativity of $\mu^\star$, this implies the saddle-point in-

equalities

$$\mathcal{L}^c(x^\star, \mu) \leq \mathcal{L}^c(x^\star, \mu^\star) \leq \mathcal{L}^c(x, \mu^\star),$$

for all $\mu \geq 0$ and all $x \in \mathcal{F}^=$. By Theorem 2.1, $x^\star$ is therefore an optimal solution of (2.60). $\square$

We now illustrate Theorem 2.4 through a simple example, highlighting how optimality is characterized via the normal cone to the base feasible set $\mathcal{F}^=$ even when classical KKT conditions fail.

Example 2.5 (Singleton Feasible Set)**.** Consider the convex optimization problem

$$\min_{x \in \mathbb{R}} f(x) := -x \quad \text{s.t.} \quad g(x) := x^2 \leq 0.$$

The feasible set is $\mathcal{F} = \{0\}$, and the unique feasible point is $x^\star = 0$. Since $g(x) = 0$ for all $x \in \mathcal{F}$, the inequality constraint acts as an equality, so $\mathcal{Q}^= = \{1\}$ and $\mathcal{Q}^< = \varnothing$. Consequently,

$$\mathcal{F}^= = \{0\}.$$

The normal cone to $\mathcal{F}^=$ at $x^\star$ is

$$N_{\mathcal{F}^=}(0) = \mathbb{R}.$$

Moreover,

$$\nabla f(0) = -1.$$

Since $\mathcal{Q}^< = \varnothing$, the first-order optimality condition (2.13) reduces to

$$-\nabla f(0) = 1 \in N_{\mathcal{F}^=}(0),$$

which holds. Therefore, $x^\star = 0$ is an optimal solution.

This example demonstrates that although no Lagrange multiplier can satisfy the classical KKT stationarity condition, the normal-cone characterization of Theorem 2.4 correctly certifies optimality.

Definition 2.6 (Cone of Directions of Constancy)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function and let $x^\star \in \mathbb{R}^n$. The *cone of directions of constancy* of f at $x^\star$ is defined as

$$\mathcal{D}_f(x^\star) := \{d \in \mathbb{R}^n \mid f(x^\star + \alpha d) = f(x^\star) \text{ for all } 0 < \alpha < \alpha_0 \text{ for some } \alpha_0 > 0\}.$$

Proposition 2.7 (Constancy–Tangent Cone Identity)**.** *Let $\mathcal{F}^{=}$ be the base feasible set induced by equality constraints and carrier inequalities, and let $x^\star \in \mathcal{F}^{=}$ be feasible. Then*

$$T_{\mathcal{F}^{=}}(x^\star) = \bigcap_{j \in \mathcal{Q}^{=}} \mathcal{D}_j(x^\star). \tag{2.15}$$

Consequently,

$$N_{\mathcal{F}^{=}}(x^\star) = \left(\bigcap_{j \in \mathcal{Q}^{=}} \mathcal{D}_j(x^\star) \right)^{+}, \tag{2.16}$$

where $(\cdot)^+$ denotes the polar cone.

Definition 2.8 (Cone of Directions of Constancy)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function and let $x^\star \in \mathbb{R}^n$. The *cone of directions of constancy* of f at $x^\star$ is defined as

$$\mathcal{D}_j(x^\star) := \left\{ d \in \mathbb{R}^n \;\middle|\; g_j(x^\star + \alpha d) = g_j(x^\star) \text{ for all } 0 < \alpha < \alpha_0 \text{ for some } \alpha_0 > 0 \right\}.$$

Proposition 2.9 (Constancy–Tangent Cone Identity)**.** *Let $\mathcal{F}^{=}$ be the base feasible set induced by equality constraints and carrier inequalities, and let $x^\star \in \mathcal{F}^{=}$. Then*

$$T_{\mathcal{F}^{=}}(x^\star) = \bigcap_{j \in \mathcal{Q}^{=}} \mathcal{D}_j(x^\star). \tag{2.17}$$

Consequently,

$$N_{\mathcal{F}^{=}}(x^\star) = \left(\bigcap_{j \in \mathcal{Q}^{=}} \mathcal{D}_j(x^\star) \right)^{+}, \tag{2.18}$$

where $(\cdot)^+$ denotes the polar cone.

Proof. We prove the two inclusions separately.

Step 1. We first show that $T_{\mathcal{F}^{=}}(x^\star) \subseteq \bigcap_{j \in \mathcal{Q}^{=}} \mathcal{D}_j(x^\star)$.

Let $d \in T_{\mathcal{F}^{=}}(x^\star)$. By definition of the tangent (contingent) cone, there exist sequences $\varepsilon_k \downarrow 0$ and $d_k \to d$ such that $x^\star + \varepsilon_k d_k \in \mathcal{F}^{=}$ for all k.

Since $x^\star + \varepsilon_k d_k \in \mathcal{F}^{=}$, we have $g_j(x^\star + \varepsilon_k d_k) = 0$ for all $j \in \mathcal{Q}^{=}$ and all k. Because $g_j(x^\star) = 0$ as well, it follows that $g_j(x^\star + \varepsilon_k d_k) = g_j(x^\star)$.

Passing to the limit and using continuity of g_j, we obtain $g_j(x^\star + \alpha d) = g_j(x^\star)$ for all sufficiently small $\alpha > 0$. Hence $d \in \mathcal{D}_j(x^\star)$ for all $j \in \mathcal{Q}^{=}$.

Step 2. We next show that $\bigcap_{j \in \mathcal{Q}^{=}} \mathcal{D}_j(x^\star) \subseteq T_{\mathcal{F}^{=}}(x^\star)$.

Let $d \in \bigcap_{j \in \mathcal{Q}^=} \mathcal{D}_j(x^\star)$. By definition, for each $j \in \mathcal{Q}^=$ there exists $\alpha_j > 0$ such that $g_j(x^\star + \alpha d) = g_j(x^\star)$ for all $\alpha \in (0, \alpha_j)$. Define $\bar{\alpha} := \min_{j \in \mathcal{Q}^=} \alpha_j$.

Then for all $\alpha \in (0, \bar{\alpha})$, we have $g_j(x^\star + \alpha d) = 0$ for all $j \in \mathcal{Q}^=$. Therefore $x^\star + \alpha d \in \mathcal{F}^=$ for all $\alpha \in (0, \bar{\alpha})$, which implies $d \in T_{\mathcal{F}^=}(x^\star)$.

Step 3. Combining the two inclusions yields $T_{\mathcal{F}^=}(x^\star) = \bigcap_{j \in \mathcal{Q}^=} \mathcal{D}_j(x^\star)$.

The normal cone identity follows immediately from the definition $N_{\mathcal{F}^=}(x^\star) = \big(T_{\mathcal{F}^=}(x^\star)\big)^+$.

□

Example 2.10 (Linear Equality Constraint)**.** Consider the convex function $g(x) = a^\top x - b$, where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$, and the equality-constrained feasible set $\mathcal{F}^= = \{x \in \mathbb{R}^n \mid a^\top x = b\}$.

Let $x^\star \in \mathcal{F}^=$ be arbitrary. For any direction $d \in \mathbb{R}^n$ and any scalar α, we have $g(x^\star + \alpha d) = b + \alpha a^\top d$. Hence $g(x^\star + \alpha d) = g(x^\star)$ for all α if and only if $a^\top d = 0$.

Therefore, the cone of directions of constancy is $\mathcal{D}_g(x^\star) = \{d \in \mathbb{R}^n \mid a^\top d = 0\}$. Since $\mathcal{F}^=$ is an affine set, its tangent cone at $x^\star$ is $T_{\mathcal{F}^=}(x^\star) = \{d \in \mathbb{R}^n \mid a^\top d = 0\}$. Thus $T_{\mathcal{F}^=}(x^\star) = \mathcal{D}_g(x^\star)$, which verifies the constancy–tangent cone identity in the affine case.

Example 2.11 (Degenerate Convex Equality)**.** Consider the convex function

$$g(x) = x^2, \quad x \in \mathbb{R},$$

with equality constraint $g(x) = 0$. The feasible set is

$$\mathcal{F}^= = \{0\}.$$

Let $x^\star = 0$. For any direction $d \in \mathbb{R}$,

$$g(0 + \alpha d) = \alpha^2 d^2.$$

Hence

$$g(0 + \alpha d) = g(0) \text{ for all sufficiently small } \alpha > 0 \iff d = 0.$$

Therefore,

$$\mathcal{D}_g(0) = \{0\}.$$

Since $\mathcal{F}^{=}$ is a singleton, its tangent cone is

$$T_{\mathcal{F}^{=}}(0) = \{0\}.$$

Thus,

$$T_{\mathcal{F}^{=}}(0) = \mathcal{D}_g(0),$$

illustrating how degeneracy collapses both the constancy and tangent cones.

Example 2.12 (Degenerate Convex Equality)**.** Consider the convex function $g(x) = x^2$ for $x \in \mathbb{R}$, with the equality constraint $g(x) = 0$. The feasible set is $\mathcal{F}^{=} = \{0\}$. Let $x^\star = 0$. For any direction $d \in \mathbb{R}$ and any scalar α, we have $g(0 + \alpha d) = \alpha^2 d^2$. Hence $g(0 + \alpha d) = g(0)$ for all sufficiently small $\alpha > 0$ if and only if $d = 0$. Therefore $\mathcal{D}_g(0) = \{0\}$. Since $\mathcal{F}^{=}$ is a singleton, its tangent cone is $T_{\mathcal{F}^{=}}(0) = \{0\}$. Thus $T_{\mathcal{F}^{=}}(0) = \mathcal{D}_g(0)$, illustrating how degeneracy collapses both the constancy and tangent cones.

2.2 Complete Lagrangian Characterization

Consider the convex optimization problem (2.60). We define the *complete Lagrangian* as

$$\mathcal{L}(x, \mu, \nu) \;:=\; f(x) \;+\; \sum_{j \in \mathcal{Q}} \mu_j g_j(x) \;+\; \sum_{i \in \mathcal{P}} \nu_i h_i(x), \tag{2.19}$$

where

$$\mu \in \mathbb{R}_+^{|\mathcal{Q}|}, \qquad \nu \in \mathbb{R}^{|\mathcal{P}|}.$$

No sign restriction is imposed on the equality multipliers ν, reflecting the two-sided nature of equality constraints.

Recall that the inequality constraint index set admits a decomposition

$$\mathcal{Q} = \mathcal{Q}^{=} \dot{\cup} \mathcal{Q}^{<},$$

and that the base feasible set is defined as

$$\mathcal{F}^{=} := \{x \mid h_i(x) = 0,\ i \in \mathcal{P},\ g_j(x) = 0,\ j \in \mathcal{Q}^{=}\}.$$

By Theorem 2.4, a feasible point $x^\star \in \mathcal{F}^{=}$ is an optimal solution of (2.60) if and only if there

exists a multiplier vector $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that

$$-\nabla f(x^\star) - \sum_{j \in \mathcal{Q}^<} \mu_j^\star \nabla g_j(x^\star) \ \in \ N_{\mathcal{F}^=}(x^\star). \tag{2.20}$$

When the inequality constraints satisfy Slater's condition, no inequality constraint can be active at every feasible point; equivalently,

$$\mathcal{Q}^= = \varnothing.$$

In this case, the base feasible set is determined solely by the equality constraints. Throughout this subsection, we further assume that the equality-feasible set

$$\{x \mid h_i(x) = 0, \ i \in \mathcal{P}\}$$

is convex.

Under this assumption, and assuming differentiability of the equality constraints, the gradients of the equality constraints generate normal directions to the feasible set in the sense that

$$\operatorname{span}\{\nabla h_i(x^\star)\}_{i \in \mathcal{P}} \ \subseteq \ N_{\mathcal{F}^=}(x^\star). \tag{2.21}$$

Consequently, if there exists a vector $\nu^\star \in \mathbb{R}^{|\mathcal{P}|}$ such that

$$\nabla f(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j^\star \nabla g_j(x^\star) + \sum_{i \in \mathcal{P}} \nu_i^\star \nabla h_i(x^\star) = 0, \tag{2.22}$$

then the normal-cone optimality condition (2.20) is satisfied. Equivalently, the Lagrangian stationarity condition

$$\nabla_x \mathcal{L}(x^\star, \mu^\star, \nu^\star) = 0 \tag{2.23}$$

is a sufficient condition for optimality.

Furthermore, if the gradients $\{\nabla h_i(x^\star)\}_{i \in \mathcal{P}}$ are linearly independent, then the normal cone to the equality-feasible set admits the exact representation

$$N_{\mathcal{F}^=}(x^\star) = \operatorname{span}\{\nabla h_i(x^\star)\}_{i \in \mathcal{P}}.$$

This condition is known as the *Linear Independence Constraint Qualification* (LICQ).

Under LICQ, the normal-cone optimality condition (2.20) is equivalent to the complete Lagrangian stationarity condition (2.23), and the equality multipliers $\nu^\star$ are unique.

Lemma 2.13 (Normal Cone to a Convex Equality-Feasible Set)**.** *Let $h_i : \mathbb{R}^n \to \mathbb{R}$, $i \in \mathcal{P}$, be differentiable functions, and define the equality-feasible set*

$$\mathcal{F}^= := \{x \mid h_i(x) = 0,\ i \in \mathcal{P}\}.$$

Assume that $\mathcal{F}^=$ is convex and let $x^\star \in \mathcal{F}^=$. Then:

1. *For each $i \in \mathcal{P}$, $\nabla h_i(x^\star) \in N_{\mathcal{F}^=}(x^\star)$. Consequently,*

$$\operatorname{span}\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}} \ \subseteq\ N_{\mathcal{F}^=}(x^\star).$$

2. *If, in addition, the gradients $\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$ are linearly independent (LICQ), then*

$$N_{\mathcal{F}^=}(x^\star) = \operatorname{span}\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}.$$

Proof. We recall that for a closed convex set C, the (convex-analytic) normal cone at $x^\star \in C$ is defined as

$$N_C(x^\star) := \{v \mid \langle v, x - x^\star\rangle \le 0,\ \forall x \in C\}.$$

Fix $i \in \mathcal{P}$. Since h_i is differentiable and convex and $h_i(x^\star) = 0$, the first-order convexity inequality yields

$$h_i(x) \ge h_i(x^\star) + \langle \nabla h_i(x^\star), x - x^\star\rangle = \langle \nabla h_i(x^\star), x - x^\star\rangle, \quad \forall x \in \mathbb{R}^n.$$

For any $x \in \mathcal{F}^=$, we have $h_i(x) = 0$, and therefore

$$\langle \nabla h_i(x^\star), x - x^\star\rangle \le 0.$$

By definition of the normal cone, this implies $\nabla h_i(x^\star) \in N_{\mathcal{F}^=}(x^\star)$. Since the normal cone is a cone and a linear subspace, it follows that

$$\operatorname{span}\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}} \subseteq N_{\mathcal{F}^=}(x^\star).$$

Suppose that the gradients $\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$ are linearly independent. Then $\mathcal{F}^=$ is a smooth

manifold near $x^\star$, and its tangent space satisfies

$$T_{\mathcal{F}^=}(x^\star) = \{d \mid \langle \nabla h_i(x^\star), d\rangle = 0,\ i \in \mathcal{P}\}.$$

Since the normal cone to a convex set coincides with the polar of the tangent cone, we obtain

$$N_{\mathcal{F}^=}(x^\star) = T_{\mathcal{F}^=}(x^\star)^\circ = \operatorname{span}\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}},$$

which proves the claim. □

Theorem 2.14 (Complete Lagrangian Characterization under Slater and LICQ)**.** *Consider the convex optimization problem (2.60). Assume that:*

- *Slater's condition holds for the inequality constraints, and*
- *the equality constraints satisfy the Linear Independence Constraint Qualification (LICQ) at $x^\star$, i.e.,*

$$\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}} \text{ are linearly independent.}$$

Suppose that all functions are differentiable.

Then, a feasible point $x^\star$ is an optimal solution of (2.60) if and only if there exist multipliers

$$\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}|}, \qquad \nu^\star \in \mathbb{R}^{|\mathcal{P}|}$$

such that the following conditions hold:

1. ***Primal feasibility:***

$$h_i(x^\star) = 0, \quad i \in \mathcal{P}, \qquad g_j(x^\star) \le 0, \quad j \in \mathcal{Q}.$$

2. ***Stationarity:***

$$\nabla_x \mathcal{L}(x^\star, \mu^\star, \nu^\star) = 0.$$

3. ***Complementary slackness:***

$$\mu_j^\star g_j(x^\star) = 0, \quad j \in \mathcal{Q}.$$

4. ***Dual feasibility:***

$$\mu_j^\star \ge 0, \quad j \in \mathcal{Q}.$$

Moreover, under LICQ the multiplier vector $\nu^\star$ is unique. Equivalently, $(x^\star, \mu^\star, \nu^\star)$ is a saddle point of the complete Lagrangian $\mathcal{L}$ over

$$\mathbb{R}^n \times \mathbb{R}_+^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}.$$

Remark 2.15 (Role of Slater's Condition and LICQ). Slater's condition implies that no inequality constraint can be active at every feasible point; that is, $\mathcal{Q}^= = \varnothing$. Consequently, all inequality constraints are effective and generate dual variables, while the base feasible set is determined solely by the equality constraints.

LICQ ensures that the normal cone to the equality-feasible set admits the exact representation $N_{\mathcal{F}^=}(x^\star) = \operatorname{span}\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}}$, which makes the normal-cone optimality condition equivalent to the complete Lagrangian stationarity condition. Under these assumptions, the generalized saddle-point and normal-cone characterizations developed earlier reduce to the classical Karush–Kuhn–Tucker conditions.

Theorem 2.14 therefore follows as a special case of the more general theory established in the absence of constraint qualifications.

2.3 Case Study: Convex Optimization with Affine Constraints under Slater's Condition

We revisit the convex optimization problem with affine equality constraints and convex inequality constraints, now assuming that Slater's condition holds for the inequality constraints. This setting illustrates how the generalized framework recovers the classical Karush–Kuhn–Tucker (KKT) conditions while preserving the geometric interpretation developed earlier.

Consider the convex optimization problem

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & Ax = b, \\ & g_j(x) \le 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{2.24}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is convex and differentiable, $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and each $g_j : \mathbb{R}^n \to \mathbb{R}$ is convex and differentiable.

The equality constraints define the base feasible set $\mathcal{F}^= := \{x \in \mathbb{R}^n \mid Ax = b\}$, which is assumed to be nonempty.

2.3.1 Slater's Condition

We assume that Slater's condition holds for the inequality constraints; that is, there exists a point $\bar{x} \in \mathcal{F}^{=}$ such that

$$g_j(\bar{x}) < 0 \quad \text{for all } j \in \mathcal{Q}.$$

Under this assumption, no inequality constraint is active everywhere on $\mathcal{F}^{=}$, and therefore

$$\mathcal{Q}^{=} = \varnothing, \qquad \mathcal{Q}^{<} = \mathcal{Q}.$$

2.3.2 Geometry of the Feasible Set

We derive explicit expressions for the tangent and normal cones associated with the affine feasible set $\mathcal{F}^{=} := \{x \in \mathbb{R}^n \mid Ax = b\}$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Tangent cone. By definition, the tangent cone to $\mathcal{F}^{=}$ at $x^\star \in \mathcal{F}^{=}$ is

$$T_{\mathcal{F}^{=}}(x^\star) := \{d \in \mathbb{R}^n \mid \exists\, \varepsilon_k \downarrow 0,\ \exists\, d_k \to d \text{ such that } x^\star + \varepsilon_k d_k \in \mathcal{F}^{=}\ \forall k\}\,.$$

For any $x^\star \in \mathcal{F}^{=}$ and any feasible sequence $x^\star + \varepsilon_k d_k \in \mathcal{F}^{=}$, we must have

$$A(x^\star + \varepsilon_k d_k) = b.$$

Since $Ax^\star = b$, this implies

$$Ad_k = 0 \quad \forall k.$$

Passing to the limit yields

$$Ad = 0.$$

Conversely, if $d \in \mathbb{R}^n$ satisfies $Ad = 0$, then

$$x^\star + \varepsilon d \in \mathcal{F}^{=} \quad \forall \varepsilon \in \mathbb{R},$$

which shows that $d \in T_{\mathcal{F}^{=}}(x^\star)$.

Therefore,

$$T_{\mathcal{F}^{=}}(x^\star) = \{d \in \mathbb{R}^n \mid Ad = 0\}.$$

Normal cone. The normal cone to $\mathcal{F}^{=}$ at $x^\star$ is defined as the polar cone of the tangent cone:

$$N_{\mathcal{F}^{=}}(x^\star) := \left\{ v \in \mathbb{R}^n \;\middle|\; v^\top d \leq 0 \quad \forall d \in T_{\mathcal{F}^{=}}(x^\star) \right\}.$$

Using the expression for the tangent cone derived above, this becomes

$$N_{\mathcal{F}^{=}}(x^\star) = \left\{ v \in \mathbb{R}^n \;\middle|\; v^\top d = 0 \quad \forall d \text{ such that } Ad = 0 \right\}.$$

The set of vectors orthogonal to $\ker(A)$ is precisely the range of $A^\top$. Hence,

$$N_{\mathcal{F}^{=}}(x^\star) = \{A^\top \lambda \mid \lambda \in \mathbb{R}^m\}.$$

For any $x^\star \in \mathcal{F}^{=}$, the tangent and normal cones are therefore given by

$$T_{\mathcal{F}^{=}}(x^\star) = \{d \in \mathbb{R}^n \mid Ad = 0\},$$
$$N_{\mathcal{F}^{=}}(x^\star) = \{A^\top \lambda \mid \lambda \in \mathbb{R}^m\}.$$

2.3.3 First-Order Optimality

By Theorem 2.4, a feasible point $x^\star \in \mathcal{F}^{=}$ is optimal for (2.24) if and only if there exists a multiplier vector $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}|}$ such that

$$-\nabla f(x^\star) - \sum_{j \in \mathcal{Q}} \mu_j^\star \nabla g_j(x^\star) \;\in\; N_{\mathcal{F}^{=}}(x^\star). \tag{2.25}$$

Equivalently, there exists $\lambda^\star \in \mathbb{R}^m$ such that

$$\nabla f(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j^\star \nabla g_j(x^\star) + A^\top \lambda^\star = 0, \tag{2.26}$$

together with the complementary slackness conditions

$$\mu_j^\star \geq 0, \qquad \mu_j^\star g_j(x^\star) = 0, \quad j \in \mathcal{Q}.$$

Under Slater's condition, the normal-cone optimality condition of Theorem 2.4 reduces exactly to the classical KKT conditions. In this case, all inequality constraints are effective, and no carrier inequalities are present. The generalized framework therefore strictly extends the classical theory, recovering it as a special case when standard constraint qualifications hold.

2.3.4 Entropy Optimization with Cost and Equality Constraints

We consider entropy regularization with a linear cost under affine equality constraints, a formulation that appears in free energy minimization, information theory, and regularized optimal transport.

Let $p = (p_1, \ldots, p_n) \in \mathbb{R}^n$ denote a probability vector and $c = (c_1, \ldots, c_n) \in \mathbb{R}^n$ a cost vector. Consider the convex program

$$\begin{aligned} \min_{p \in \mathbb{R}^n} \quad & f(p) := \sum_{i=1}^n p_i \log p_i + \sum_{i=1}^n c_i p_i \\ \text{s.t.} \quad & \sum_{i=1}^n p_i = 1, \\ & \sum_{i=1}^n a_i p_i = b, \\ & p_i \geq 0, \quad i = 1, \ldots, n, \end{aligned} \tag{2.27}$$

where $a_i \in \mathbb{R}$ and $b \in \mathbb{R}$ are given constants. The objective f is strictly convex on the probability simplex.

Base feasible set. The equality constraints define the affine set

$$\mathcal{F}^= = \left\{ p \in \mathbb{R}^n \;\middle|\; \sum_{i=1}^n p_i = 1, \; \sum_{i=1}^n a_i p_i = b \right\} = \{ p \mid Ap = b' \}, \tag{2.28}$$

where

$$A = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}, \qquad b' = \begin{pmatrix} 1 \\ b \end{pmatrix}.$$

For any $p^\star \in \mathcal{F}^=$, the tangent and normal cones are

$$T_{\mathcal{F}^=}(p^\star) = \left\{ d \in \mathbb{R}^n \;\middle|\; \sum_{i=1}^n d_i = 0, \; \sum_{i=1}^n a_i d_i = 0 \right\}, \tag{2.29}$$

$$N_{\mathcal{F}^=}(p^\star) = \{ \lambda_1 \mathbf{1} + \lambda_2 a \mid \lambda_1, \lambda_2 \in \mathbb{R} \}, \tag{2.30}$$

with $\mathbf{1} = (1, \ldots, 1)$ and $a = (a_1, \ldots, a_n)$.

First-order optimality. The gradient of the objective is

$$\nabla f(p)_i = 1 + \log p_i + c_i. \tag{2.31}$$

By Theorem 2.4, a feasible point $p^\star \in \mathcal{F}^=$ is optimal for (2.27) if and only if

$$-\nabla f(p^\star) \in N_{\mathcal{F}^=}(p^\star). \tag{2.32}$$

Using (2.30), this condition is equivalent to the existence of $\lambda_1, \lambda_2 \in \mathbb{R}$ such that

$$-(1 + \log p_i^\star + c_i) = \lambda_1 + \lambda_2 a_i, \quad i = 1, \ldots, n.$$

Solving yields

$$p_i^\star = \frac{e^{-c_i - \lambda_2 a_i}}{\sum_{k=1}^n e^{-c_k - \lambda_2 a_k}}, \qquad i = 1, \ldots, n, \tag{2.33}$$

where λ_2 is chosen so that $\sum_i a_i p_i^\star = b$.

Proposition 2.16 (Entropy–Cost Optimization under Affine Constraints)**.** *Assume that the feasible set $\mathcal{F}^=$ defined in (2.28) is nonempty. Then the convex program (2.27) admits a unique optimal solution $p^\star$, which is characterized by the Gibbs–Boltzmann form (2.33). Conversely, any $p^\star$ of the form (2.33) satisfying the moment constraint $\sum_i a_i p_i^\star = b$ is the unique optimal solution of (2.27).*

2.4 Associated Dual Problems: Restricted and Complete Lagrangian Formulations

This subsection presents two dual formulations associated with the convex optimization problem (2.60). The first is a *restricted dual formulation*, which remains valid in the absence of Slater's condition. The second is the classical *complete dual formulation*, which applies under Slater's condition and the Linear Independence Constraint Qualification (LICQ). The two formulations coincide when strict feasibility holds, but differ substantially when it does not.

2.4.1 Restricted Lagrangian Formulation

Let $\mathcal{Q}^<$ denote the index set of *effective* inequality constraints, and let $\mathcal{Q}^=$ denote the complementary index set of ineffective inequality constraints. Define the *base feasible set*

$$\mathcal{F}^= := \{x \in \mathbb{R}^n : h_i(x) = 0,\ i \in \mathcal{P},\ g_j(x) = 0,\ j \in \mathcal{Q}^=\},$$

obtained by incorporating all equality constraints and the inequality constraints indexed by $\mathcal{Q}^=$ directly into the feasible region.

Definition 2.17 (Restricted Lagrangian)**.** The *restricted Lagrangian* associated with (2.60) is defined by

$$\mathcal{L}^c(x,\mu) = f(x) + \sum_{j\in\mathcal{Q}^<} \mu_j g_j(x), \qquad x \in \mathcal{F}^=,\ \mu \in \mathbb{R}^c_+. \tag{2.34}$$

Unlike the complete Lagrangian, multipliers are introduced only for the inequality constraints indexed by $\mathcal{Q}^<$. Constraints indexed by $\mathcal{Q}^=$ are enforced through the geometry of $\mathcal{F}^=$ and do not give rise to independent dual variables.

Proposition 2.18 (Restricted Dual Function)**.** *The restricted dual function associated with $\mathcal{L}^c$ is given by*

$$d(\mu) := \inf_{x\in\mathcal{F}^=} \left\{ f(x) + \sum_{j\in\mathcal{Q}^<} \mu_j g_j(x) \right\}, \qquad \mu \in \mathbb{R}^c_+. \tag{2.35}$$

Moreover, $d(\mu)$ is a concave function on $\mathbb{R}^c_+$.

Proof. For each fixed $\mu \in \mathbb{R}^c_+$, the function $x \mapsto f(x) + \sum_{j\in\mathcal{Q}^<} \mu_j g_j(x)$ is convex, since f and g_j are convex. Because $\mathcal{F}^=$ is a convex set, the infimum over $\mathcal{F}^=$ defines a concave function of μ. □

Definition 2.19 (Restricted Dual Problem)**.** The *restricted dual problem* associated with (2.60) is

$$\max_{\mu\in\mathbb{R}^c_+} \quad d(\mu), \tag{2.36}$$

where $d(\mu)$ is defined in (2.35).

Theorem 2.20 (Weak Duality for the Restricted Formulation)**.** *For any primal feasible point $x \in \mathcal{F}$ and any $\mu \in \mathbb{R}^c_+$,*

$$d(\mu) \le \mathcal{L}^c(x,\mu) \le f(x).$$

Consequently, the optimal value of the restricted dual problem provides a lower bound on the primal optimal value.

Proof. By definition of $d(\mu)$,

$$d(\mu) = \inf_{y\in\mathcal{F}^=} \mathcal{L}^c(y,\mu) \le \mathcal{L}^c(x,\mu)$$

for any $x \in \mathcal{F}^=$. If $x \in \mathcal{F}$ is primal feasible, then $g_j(x) \le 0$ for all $j \in \mathcal{Q}^<$, and hence

$$\mathcal{L}^c(x,\mu) = f(x) + \sum_{j\in\mathcal{Q}^<} \mu_j g_j(x) \le f(x),$$

since $\mu_j \geq 0$. $\square$

Theorem 2.21 (Strong Duality via Restricted Saddle Points)**.** *Strong duality holds between the primal problem (2.60) and the restricted dual problem (2.36) if and only if there exists $(x^\star, \mu^\star) \in \mathcal{F}^= \times \mathbb{R}^c_+$ such that $(x^\star, \mu^\star)$ is a saddle point of the restricted Lagrangian $\mathcal{L}^c$, i.e.,*

$$\mathcal{L}^c(x^\star, \mu) \leq \mathcal{L}^c(x^\star, \mu^\star) \leq \mathcal{L}^c(x, \mu^\star), \quad \forall x \in \mathcal{F}^=, \forall \mu \in \mathbb{R}^c_+. \tag{2.37}$$

In this case,

$$\min_{x \in \mathcal{F}} f(x) = \max_{\mu \in \mathbb{R}^c_+} d(\mu),$$

and $\mu^\star$ is dual optimal.

Proof. If $(x^\star, \mu^\star)$ is a saddle point of $\mathcal{L}^c$, then

$$d(\mu^\star) = \inf_{x \in \mathcal{F}^=} \mathcal{L}^c(x, \mu^\star) = \mathcal{L}^c(x^\star, \mu^\star).$$

By the right-hand inequality in (2.37) and primal feasibility of $x^\star$,

$$\mathcal{L}^c(x^\star, \mu^\star) \leq f(x^\star).$$

Combining this with weak duality yields

$$d(\mu^\star) = f(x^\star),$$

which establishes strong duality and dual optimality of $\mu^\star$.

Conversely, if strong duality holds and $f(x^\star) = d(\mu^\star)$, then $(x^\star, \mu^\star)$ satisfies the saddle-point inequalities (2.37) by standard convex duality arguments. $\square$

2.4.2 Complete Lagrangian Formulation

We now consider the classical Lagrangian formulation under standard regularity assumptions.

Assumption 2.22 (Slater and LICQ)**.** Assume that:

- Slater's condition holds for the inequality constraints, i.e., there exists $\bar{x} \in \mathbb{R}^n$ such that

$$g_j(\bar{x}) < 0, \quad \forall j \in \mathcal{Q}, \qquad h_i(\bar{x}) = 0, \quad \forall i \in \mathcal{P};$$

- the equality constraints satisfy the Linear Independence Constraint Qualification (LICQ) at an optimal solution $x^\star$, i.e.,

$$\{\nabla h_i(x^\star)\}_{i\in\mathcal{P}} \text{ are linearly independent.}$$

Definition 2.23 (Complete Lagrangian)**.** Under Assumption 2.22, the *complete Lagrangian* associated with the convex optimization problem (2.60) is defined as

$$\mathcal{L}(x,\mu,\nu) = f(x) + \sum_{j\in\mathcal{Q}} \mu_j g_j(x) + \sum_{i\in\mathcal{P}} \nu_i h_i(x), \qquad \mu \in \mathbb{R}_+^{|\mathcal{Q}|},\ \nu \in \mathbb{R}^{|\mathcal{P}|}. \tag{2.38}$$

Proposition 2.24 (Complete Dual Function)**.** *The dual function associated with the complete Lagrangian is given by*

$$d(\mu,\nu) := \inf_{x\in\mathbb{R}^n} \left\{ f(x) + \sum_{j\in\mathcal{Q}} \mu_j g_j(x) + \sum_{i\in\mathcal{P}} \nu_i h_i(x) \right\}. \tag{2.39}$$

Moreover, $d(\mu,\nu)$ is concave on $\mathbb{R}_+^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

Proof. For each fixed (μ,ν) with $\mu \geq 0$, the mapping

$$x \mapsto f(x) + \sum_{j\in\mathcal{Q}} \mu_j g_j(x) + \sum_{i\in\mathcal{P}} \nu_i h_i(x)$$

is convex, since f and g_j are convex and h_i are affine. Taking the infimum over $x \in \mathbb{R}^n$ therefore yields a concave function of (μ,ν). □

Definition 2.25 (Complete Dual Problem)**.** The *complete dual problem* associated with (2.60) is

$$\begin{aligned} \max_{\mu,\nu} \quad & d(\mu,\nu) \\ \text{s.t.} \quad & \mu \geq 0, \end{aligned} \tag{2.40}$$

where $d(\mu,\nu)$ is defined in (2.39).

Theorem 2.26 (Weak Duality for the Complete Formulation)**.** *For any primal feasible point $x \in \mathcal{F}$ and any dual feasible (μ,ν),*

$$d(\mu,\nu) \leq \mathcal{L}(x,\mu,\nu) \leq f(x).$$

Proof. By definition of $d(\mu, \nu)$,

$$d(\mu, \nu) = \inf_{y \in \mathbb{R}^n} \mathcal{L}(y, \mu, \nu) \leq \mathcal{L}(x, \mu, \nu).$$

If x is primal feasible, then $g_j(x) \leq 0$ and $h_i(x) = 0$ for all j, i, and since $\mu \geq 0$,

$$\mathcal{L}(x, \mu, \nu) = f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x) \leq f(x).$$

□

Theorem 2.27 (Strong Duality and Saddle-Point Characterization)**.** *Under Assumption 2.22, strong duality holds between the primal problem (2.60) and the complete dual problem (2.40), i.e.,*

$$\min_{x \in \mathcal{F}} f(x) = \max_{\mu \geq 0, \nu} d(\mu, \nu). \tag{2.41}$$

Moreover, there exists a primal–dual optimal triple $(x^\star, \mu^\star, \nu^\star)$ such that

$$f(x^\star) = d(\mu^\star, \nu^\star).$$

This equality holds if and only if $(x^\star, \mu^\star, \nu^\star)$ is a saddle point of the complete Lagrangian, i.e.,

$$\mathcal{L}(x^\star, \mu, \nu) \leq \mathcal{L}(x^\star, \mu^\star, \nu^\star) \leq \mathcal{L}(x, \mu^\star, \nu^\star), \quad \forall x \in \mathbb{R}^n,\ \forall \mu \geq 0,\ \forall \nu. \tag{2.42}$$

Proof. Strong duality follows from Slater's condition for convex optimization problems. The saddle-point characterization is a standard consequence of convex–concave minimax theory and is equivalent to the KKT conditions stated in Theorem 2.14. □

Corollary 2.28 (Uniqueness of Equality Multipliers)**.** *Under LICQ, the equality multiplier vector $\nu^\star$ is unique. Consequently, the optimal value function is locally differentiable with respect to perturbations of the equality constraints.*

Proof. LICQ ensures linear independence of the gradients of the equality constraints at $x^\star$, which implies uniqueness of the corresponding Lagrange multipliers by standard sensitivity analysis results. □

Comparison with the Restricted Formulation. When Slater's condition holds, all inequality constraints are effective and the restricted and complete Lagrangian formulations coincide.

When Slater's condition fails, the complete Lagrangian formulation may fail to admit a saddle point or a dual optimizer. In such cases, the restricted Lagrangian formulation provides a dual characterization that restores strong duality by introducing multipliers only for the effective inequality constraints, while absorbing the remaining constraints into the geometry of the feasible set.

2.5 Case Study: Discrete-Time Convex Optimal Control

We illustrate how the Lagrangian duality framework developed earlier extends to dynamic optimization problems by considering a discrete-time convex optimal control problem. This case study makes explicit how strong duality, adjoint equations, and stagewise decomposition arise from the complete Lagrangian.

Problem 2.29 (Discrete-Time Convex Optimal Control)**.** Let $T \geq 2$ be a finite horizon. Consider the optimization problem

$$\begin{aligned} \min_{\{u_t\}_{t=1}^{T-1}} \quad & \sum_{t=1}^{T-1} f_t(u_t) \\ \text{s.t.} \quad & x_{t+1} = A_t x_t + B_t u_t, \quad t = 1, \ldots, T-1, \\ & x_T = d, \\ & u_t \in U_t, \quad t = 1, \ldots, T-1, \end{aligned} \tag{2.43}$$

where the initial state $x_1 \in \mathbb{R}^n$ is given. The state variables satisfy $x_t \in \mathbb{R}^n$, the control variables satisfy $u_t \in \mathbb{R}^m$, the matrices $A_t \in \mathbb{R}^{n \times n}$ and $B_t \in \mathbb{R}^{n \times m}$ are given, and $d \in \mathbb{R}^n$ is a prescribed terminal state.

Assumption 2.30 (Convexity and Slater Condition)**.** For each $t = 1, \ldots, T-1$, the function $f_t : \mathbb{R}^m \to \mathbb{R}$ is convex and the set $U_t \subset \mathbb{R}^m$ is convex. Moreover, Slater's condition holds: there exists a control sequence $\{\bar{u}_t\}_{t=1}^{T-1}$ with $\bar{u}_t \in \operatorname{int} U_t$ such that the corresponding state trajectory $\{\bar{x}_t\}_{t=1}^{T}$, generated by (2.43), satisfies $\bar{x}_T = d$.

Under Assumption 2.30, problem (2.43) is a convex optimization problem for which strong duality holds.

Complete Lagrangian. Introduce Lagrange multipliers $\mu_t \in \mathbb{R}^n$ for the state transition constraints and a multiplier $\nu \in \mathbb{R}^n$ for the terminal constraint $x_T = d$. The complete Lagrangian

associated with (2.43) is

$$\mathcal{L}(x,u,\mu,\nu) = \sum_{t=1}^{T-1} f_t(u_t) + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} - A_t x_t - B_t u_t \rangle + \langle \nu, d - x_T \rangle. \tag{2.44}$$

Dual function. The dual function is defined by minimizing the Lagrangian over the primal variables, subject to $u_t \in \mathcal{U}_t$ and with x_1 fixed:

$$d(\mu,\nu) := \inf_{\substack{x_2,\ldots,x_T \in \mathbb{R}^n \\ u_t \in \mathcal{U}_t}} \mathcal{L}(x,u,\mu,\nu). \tag{2.45}$$

Expanding the inner products in (2.44) yields

$$\mathcal{L} = \sum_{t=1}^{T-1} f_t(u_t) + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} \rangle - \sum_{t=1}^{T-1} \langle \mu_t, A_t x_t \rangle - \sum_{t=1}^{T-1} \langle \mu_t, B_t u_t \rangle + \langle \nu, d \rangle - \langle \nu, x_T \rangle.$$

Using $\langle \mu_t, A_t x_t \rangle = \langle A_t^\top \mu_t, x_t \rangle$, we group terms according to the state variables.

The terminal state x_T appears only through

$$\langle \mu_{T-1}, x_T \rangle - \langle \nu, x_T \rangle = \langle \mu_{T-1} - \nu, x_T \rangle.$$

If $\mu_{T-1} \neq \nu$, the infimum over $x_T \in \mathbb{R}^n$ is $-\infty$. Hence, finiteness of the dual function requires

$$\mu_{T-1} = \nu. \tag{2.46}$$

For intermediate states x_t, $t = 2, \ldots, T-1$, the corresponding terms are

$$\langle \mu_{t-1}, x_t \rangle - \langle A_t^\top \mu_t, x_t \rangle = \langle \mu_{t-1} - A_t^\top \mu_t, x_t \rangle.$$

Again, finiteness of the infimum over $x_t \in \mathbb{R}^n$ requires

$$\mu_{t-1} = A_t^\top \mu_t, \qquad t = 2, \ldots, T-1. \tag{2.47}$$

Equations (2.46) and (2.47) together yield the discrete-time adjoint equations

$$\mu_{t-1} = A_t^\top \mu_t, \quad t = 2, \ldots, T-1, \qquad \mu_{T-1} = \nu. \tag{2.48}$$

Assuming (2.48) holds, all dependence on $x_2, \ldots, x_T$ vanishes and the dual function reduces

to

$$d(\mu,\nu) = \langle \nu, d\rangle - \langle A_1^\top \mu_1, x_1\rangle + \sum_{t=1}^{T-1} \left(f_t(u_t) - \langle \mu_t, B_t u_t\rangle\right), \tag{2.49}$$

where the term involving x_1 is constant.

The control variables now decouple across time. Minimization over $u_t \in U_t$ therefore separates as

$$\inf_{u_t \in U_t} \sum_{t=1}^{T-1} \left(f_t(u_t) - \langle \mu_t, B_t u_t\rangle\right) = \sum_{t=1}^{T-1} \inf_{u_t \in U_t} \left(f_t(u_t) - \langle \mu_t, B_t u_t\rangle\right).$$

For each t, the stagewise minimization problem is

$$\min_{u_t \in U_t} \left\{f_t(u_t) - \langle \mu_t, B_t u_t\rangle\right\}. \tag{2.50}$$

Let $\hat{u}_t(\mu_t)$ denote its (assumed unique) minimizer.

Substituting $\hat{u}_t(\mu_t)$ into (2.49) yields the dual function

$$d(\nu) = \sum_{t=1}^{T-1} \left[f_t\big(\hat{u}_t(\mu_t(\nu))\big) - \langle \mu_t(\nu), B_t \hat{u}_t(\mu_t(\nu))\rangle\right] + \langle \nu, d\rangle - \langle A_1^\top \mu_1(\nu), x_1\rangle, \tag{2.51}$$

where $\mu_t(\nu)$ is determined uniquely from ν via (2.48).

The associated dual problem is

$$\max_{\nu \in \mathbb{R}^n} d(\nu). \tag{2.52}$$

Theorem 2.31 (Dynamic Decomposition via the Complete Lagrangian)**.** *Under Assumption 2.30, the following statements hold:*

1. *Strong duality holds between the primal problem (2.43) and the dual problem (2.52).*
2. *There exist multipliers $(\mu^\star, \nu^\star)$ satisfying the adjoint equations (2.48).*
3. *The optimal control is obtained stagewise by solving (2.50) with $\mu_t = \mu_t^\star$.*
4. *A feasible trajectory $(x^\star, u^\star)$ is optimal if and only if $(x^\star, u^\star, \mu^\star, \nu^\star)$ is a saddle point of the complete Lagrangian (2.44).*

2.6 Extension to Output-Feedback Linear–Quadratic Regulator

We now extend the dynamic duality framework to a linear–quadratic regulator (LQR) with output penalties. This extension illustrates how quadratic state and output costs modify the adjoint equations and lead naturally to the Riccati recursion through the dual formulation.

Primal problem with output penalty. Let the system dynamics be

$$x_{t+1} = A_t x_t + B_t u_t, \qquad y_t = C_t x_t,$$

and consider the finite-horizon optimization problem

$$\begin{aligned} \min_{\{u_t\}_{t=1}^{T-1}} \quad & \sum_{t=1}^{T-1} \left(\tfrac{1}{2} u_t^\top R_t u_t + \tfrac{1}{2} y_t^\top Q_t y_t \right) \\ \text{s.t.} \quad & x_{t+1} = A_t x_t + B_t u_t, \\ & x_T = d, \end{aligned} \tag{2.53}$$

where $R_t \in \mathbb{R}^{m \times m}$ satisfies $R_t \succ 0$ and $Q_t \in \mathbb{R}^{p \times p}$ satisfies $Q_t \succeq 0$. Equivalently, the stage cost can be written as

$$\tfrac{1}{2} u_t^\top R_t u_t + \tfrac{1}{2} x_t^\top C_t^\top Q_t C_t x_t.$$

Problem (2.53) is a strictly convex optimization problem, and strong duality holds.

Complete Lagrangian. Introduce multipliers $\mu_t \in \mathbb{R}^n$ for the state transition constraints and $\nu \in \mathbb{R}^n$ for the terminal constraint $x_T = d$. The complete Lagrangian is

$$\begin{aligned} \mathcal{L}(x, u, \mu, \nu) = & \sum_{t=1}^{T-1} \left(\tfrac{1}{2} u_t^\top R_t u_t + \tfrac{1}{2} x_t^\top C_t^\top Q_t C_t x_t \right) \\ & + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} - A_t x_t - B_t u_t \rangle + \langle \nu, d - x_T \rangle. \end{aligned} \tag{2.54}$$

Minimization with respect to the state variables. Collecting terms involving x_t for $t = 2, \ldots, T-1$ yields

$$\tfrac{1}{2} x_t^\top C_t^\top Q_t C_t x_t + \langle \mu_{t-1}, x_t \rangle - \langle A_t^\top \mu_t, x_t \rangle.$$

Minimization over $x_t \in \mathbb{R}^n$ requires the first-order condition

$$C_t^\top Q_t C_t x_t + \mu_{t-1} - A_t^\top \mu_t = 0,$$

which gives the modified adjoint recursion

$$\mu_{t-1} = A_t^\top \mu_t - C_t^\top Q_t C_t x_t, \qquad t = 2, \ldots, T-1. \tag{2.55}$$

For the terminal state x_T, as before, finiteness of the dual function implies

$$\mu_{T-1} = \nu.$$

Minimization with respect to the control variables. The terms involving u_t are

$$\tfrac{1}{2} u_t^\top R_t u_t - \langle \mu_t, B_t u_t \rangle.$$

Taking the gradient with respect to u_t yields

$$R_t u_t - B_t^\top \mu_t = 0,$$

and therefore the optimal control is

$$u_t^\star = R_t^{-1} B_t^\top \mu_t. \tag{2.56}$$

Riccati recursion. At optimality, the costate is linear in the state, i.e.,

$$\mu_t = P_{t+1} x_{t+1}.$$

Substituting this relation, together with the dynamics and (2.56), into (2.55) yields the backward Riccati recursion

$$\begin{aligned} P_t = {} & C_t^\top Q_t C_t + A_t^\top P_{t+1} A_t \\ & - A_t^\top P_{t+1} B_t \left(R_t + B_t^\top P_{t+1} B_t \right)^{-1} B_t^\top P_{t+1} A_t, \end{aligned} \tag{2.57}$$

with terminal condition $P_T = 0$ (or $P_T = C_T^\top Q_T C_T$ if a terminal output penalty is included).

Optimal feedback law. The resulting optimal control policy is the state-feedback law

$$u_t^\star = - \left(R_t + B_t^\top P_{t+1} B_t \right)^{-1} B_t^\top P_{t+1} A_t x_t^\star. \tag{2.58}$$

Theorem 2.32 (Optimal Control and Riccati Characterization for Output-Weighted LQR)**.** *Consider the output-weighted linear–quadratic regulator problem (2.53). There exists a unique sequence of symmetric matrices $\{P_t\}_{t=1}^T$, $P_t \succeq 0$, satisfying the backward Riccati recursion (2.57), with terminal condition $P_T = 0$ (or $P_T = C_T^\top Q_T C_T$ if a terminal output penalty is included).*

Furthermore, the optimal control sequence $\{u_t^\star\}_{t=1}^{T-1}$ is given by the state-feedback law (2.58). The

associated costate variables satisfy $\mu_t = P_{t+1}x^\star_{t+1}$, and are consistent with the adjoint equations (2.55). The optimal solution of (2.53) is completely characterized by the Riccati equation (2.57) and the feedback law (2.58).

Theorem 2.33 (Dual Optimal Control Formulation of LQR)**.** *The Lagrangian dual of the output-feedback LQR problem (2.53) is the following concave maximization problem:*

$$\begin{aligned} \max_{\{\mu_t\}_{t=1}^{T-1},\nu} \quad & \langle \nu, d\rangle - \langle \mu_1, A_1 x_1\rangle - \sum_{t=1}^{T-1} \tfrac{1}{2}\mu_t^\top B_t R_t^{-1} B_t^\top \mu_t \\ s.t. \quad & \mu_{t-1} = A_t^\top \mu_t - C_t^\top Q_t C_t x_t, \quad t = 2,\dots,T-1, \\ & \mu_{T-1} = \nu, \end{aligned} \tag{2.59}$$

where the state trajectory $\{x_t\}$ is implicitly defined by the adjoint recursion. Strong duality holds, and the optimal primal control is recovered via

$$u_t^\star = R_t^{-1} B_t^\top \mu_t^\star.$$

Proof. We derive the dual problem by explicit minimization of the complete Lagrangian associated with the output-weighted LQR problem (2.53).

Recall that the complete Lagrangian is

$$\mathcal{L}(x,u,\mu,\nu) = \sum_{t=1}^{T-1}\left(\tfrac{1}{2}u_t^\top R_t u_t + \tfrac{1}{2}x_t^\top C_t^\top Q_t C_t x_t\right) + \sum_{t=1}^{T-1}\langle \mu_t, x_{t+1} - A_t x_t - B_t u_t\rangle + \langle \nu, d - x_T\rangle,$$

where x_1 is fixed.

The dual function is defined as

$$d(\mu,\nu) := \inf_{\substack{x_2,\dots,x_T\in\mathbb{R}^n \\ u_t\in\mathbb{R}^m}} \mathcal{L}(x,u,\mu,\nu).$$

Expanding the inner products and grouping terms according to the state variables, the dependence on the terminal state x_T appears only through

$$\langle \mu_{T-1} - \nu, x_T\rangle.$$

Since this expression is affine in x_T, the infimum over $x_T \in \mathbb{R}^n$ is finite if and only if

$$\mu_{T-1} = \nu.$$

For each intermediate state x_t, $t = 2, \ldots, T-1$, the corresponding terms are

$$\tfrac{1}{2} x_t^\top C_t^\top Q_t C_t x_t + \langle \mu_{t-1} - A_t^\top \mu_t, x_t \rangle.$$

This expression is strictly convex in x_t. Its infimum is finite and attained if and only if the first-order optimality condition holds, yielding

$$\mu_{t-1} = A_t^\top \mu_t - C_t^\top Q_t C_t x_t, \qquad t = 2, \ldots, T-1.$$

Together with $\mu_{T-1} = \nu$, these relations define the adjoint (costate) recursion in (2.59). When these conditions are satisfied, all dependence on $x_2, \ldots, x_T$ is eliminated from the dual function.

The remaining terms involving the control variables are

$$\tfrac{1}{2} u_t^\top R_t u_t - \langle \mu_t, B_t u_t \rangle, \qquad t = 1, \ldots, T-1.$$

Since $R_t \succ 0$, each term is strictly convex in u_t and admits a unique minimizer. The first-order optimality condition gives

$$R_t u_t = B_t^\top \mu_t, \qquad u_t^\star = R_t^{-1} B_t^\top \mu_t.$$

Substituting $u_t^\star$ back into the Lagrangian yields

$$\inf_{u_t} \left(\tfrac{1}{2} u_t^\top R_t u_t - \langle \mu_t, B_t u_t \rangle \right) = -\tfrac{1}{2} \mu_t^\top B_t R_t^{-1} B_t^\top \mu_t.$$

Collecting all remaining terms, the dual function reduces to

$$d(\mu, \nu) = \langle \nu, d \rangle - \langle \mu_1, A_1 x_1 \rangle - \sum_{t=1}^{T-1} \tfrac{1}{2} \mu_t^\top B_t R_t^{-1} B_t^\top \mu_t,$$

provided the adjoint relations are satisfied; otherwise $d(\mu, \nu) = -\infty$. Maximizing this concave function subject to the adjoint constraints yields the dual problem (2.59).

Under the standing convexity assumptions and Slater's condition for the primal problem, strong duality holds. Hence, there exists an optimal dual solution $(\mu^\star, \nu^\star)$ attaining the primal optimal value. The corresponding primal control recovered via

$$u_t^\star = R_t^{-1} B_t^\top \mu_t^\star$$

is optimal for the primal problem. □

2.7 Parametric Value Function and Complete Lagrangian Optimality

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g_j(x) \leq \theta_j, \quad j \in \mathcal{Q}, \\ & h_i(x) = \delta_i, \quad i \in \mathcal{P}, \end{aligned} \tag{2.60}$$

where f and g_j are convex functions and h_i are affine. The vectors $\theta \in \mathbb{R}^{|\mathcal{Q}|}$ and $\delta \in \mathbb{R}^{|\mathcal{P}|}$ parameterize the right-hand sides of the inequality and equality constraints, respectively.

Rather than fixing the constraint levels a priori, we regard (2.60) as a member of a family of optimization problems indexed by the parameters (θ, δ). This parametric perspective allows us to study how the optimal cost varies as the constraints are tightened or relaxed. For given (θ, δ), define the feasible set

$$\mathcal{F}(\theta, \delta) = \{x \in \mathbb{R}^n : g(x) \leq \theta,\ h(x) = \delta\}. \tag{2.61}$$

The set $\mathcal{F}(\theta, \delta)$ collects all decision variables x that satisfy the inequality constraints with upper bounds θ and the equality constraints with right-hand side δ. When θ increases componentwise, the feasible set expands, while changes in δ translate the affine constraint manifold.

The associated *optimal value function* is defined by

$$v(\theta, \delta) = \begin{cases} \inf\limits_{x \in \mathcal{F}(\theta,\delta)} f(x), & \mathcal{F}(\theta, \delta) \neq \varnothing, \\ +\infty, & \text{otherwise.} \end{cases} \tag{2.62}$$

Thus, $v(\theta, \delta)$ represents the minimum achievable objective value under constraint levels (θ, δ). Assigning the value $+\infty$ to infeasible parameter pairs allows v to be treated as an extended-real-valued convex function on $\mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

The original optimization problem corresponds to the nominal parameter choice $(\theta, \delta) = (0, 0)$. In this sense, $v(\theta, \delta)$ provides a global description of the problem's sensitivity to perturbations in both inequality and equality constraints and serves as the central object linking dual variables, Lagrange multipliers, and first-order optimality conditions.

Lemma 2.34 (Convexity of the value function)**.** *The function*

$$v : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \to \mathbb{R} \cup \{+\infty\}$$

is convex.

Proof. Let (θ^1, δ^1) and (θ^2, δ^2) be given and $\alpha \in (0, 1)$. Set

$$(\theta, \delta) = \alpha(\theta^1, \delta^1) + (1 - \alpha)(\theta^2, \delta^2).$$

For any $x^1 \in \mathcal{F}(\theta^1, \delta^1)$ and $x^2 \in \mathcal{F}(\theta^2, \delta^2)$, convexity of g_j and affinity of h_i imply

$$g(\alpha x^1 + (1 - \alpha)x^2) \leq \theta, \qquad h(\alpha x^1 + (1 - \alpha)x^2) = \delta.$$

Hence $\alpha x^1 + (1 - \alpha)x^2 \in \mathcal{F}(\theta, \delta)$. By convexity of f,

$$f(\alpha x^1 + (1 - \alpha)x^2) \leq \alpha f(x^1) + (1 - \alpha)f(x^2).$$

Taking infima over x^1 and x^2 yields

$$v(\theta, \delta) \leq \alpha v(\theta^1, \delta^1) + (1 - \alpha)v(\theta^2, \delta^2),$$

proving convexity. □

Define the complete Lagrangian

$$\mathcal{L}(x, \mu, \nu) = f(x) + \langle \mu, g(x) - \theta \rangle + \langle \nu, h(x) - \delta \rangle, \tag{2.63}$$

with multipliers $\mu \in \mathbb{R}_+^{|\mathcal{Q}|}$ and $\nu \in \mathbb{R}^{|\mathcal{P}|}$. Fix (θ^0, δ^0) and define the extended dual function

$$D(\mu, \nu) = \begin{cases} \inf\limits_{x \in \mathbb{R}^n} \left[f(x) + \langle \mu, g(x) - \theta^0 \rangle + \langle \nu, h(x) - \delta^0 \rangle \right], & \mu \geq 0, \\ -\infty, & \text{otherwise.} \end{cases} \tag{2.64}$$

2.7.1 Fenchel conjugate of the value function.

Since the optimal value function

$$v : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \to \mathbb{R} \cup \{+\infty\}$$

is convex, we may associate with it its *Fenchel conjugate* $v^* : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \to \mathbb{R} \cup \{+\infty\}$, defined by

$$v^*(s,u) := \sup_{(\theta,\delta)} \Big\{ \langle s,\theta\rangle + \langle u,\delta\rangle - v(\theta,\delta) \Big\}, \tag{2.65}$$

where $(s,u) \in \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

Equivalently, $v^*(s,u)$ is the support function of the epigraph of v and encodes all affine lower bounds of v. In particular, (s,u) belongs to the subdifferential $\partial v(\theta^0,\delta^0)$ if and only if

$$v(\theta^0,\delta^0) + v^*(s,u) = \langle s,\theta^0\rangle + \langle u,\delta^0\rangle, \tag{2.66}$$

which is the Fenchel–Young equality.

Lemma 2.35 (Conjugate representation of the dual function)**.** *Let v^* denote the Fenchel conjugate of v. Then*

$$D(\mu,\nu) = -\,v^*(-\mu,-\nu) - \langle \mu,\theta^0\rangle - \langle \nu,\delta^0\rangle. \tag{2.67}$$

Proof. For $\mu \geq 0$ we use the identities

$$\langle \mu, g(x) - \theta^0\rangle = \inf_{\theta \geq g(x)} \langle \mu, \theta - \theta^0\rangle, \qquad \langle \nu, h(x) - \delta^0\rangle = \inf_{\delta = h(x)} \langle \nu, \delta - \delta^0\rangle.$$

Substituting and exchanging infima,

$$D(\mu,\nu) = \inf_{(\theta,\delta)} \big[v(\theta,\delta) + \langle \mu,\theta-\theta^0\rangle + \langle \nu,\delta-\delta^0\rangle \big].$$

By definition of the Fenchel conjugate,

$$\inf_{(\theta,\delta)} \big[v(\theta,\delta) + \langle \mu,\theta\rangle + \langle \nu,\delta\rangle \big] = -\,v^*(-\mu,-\nu),$$

which yields the stated formula. □

2.7.2 Subdifferential and sensitivity of the value function.

Let

$$v : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \to \mathbb{R} \cup \{+\infty\}$$

be the optimal value function associated with the parametric problem. Since v is convex (Lemma 2.34), it admits a *subdifferential* at every point.

Definition 2.36 (Subdifferential)**.** The *subdifferential* of v at (θ^0, δ^0), denoted $\partial v(\theta^0, \delta^0)$, is the set of all vectors $(s, u) \in \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$ such that

$$v(\theta, \delta) \geq v(\theta^0, \delta^0) + \langle s, \theta - \theta^0 \rangle + \langle u, \delta - \delta^0 \rangle \quad \text{for all } (\theta, \delta). \tag{2.68}$$

Each element of $\partial v(\theta^0, \delta^0)$ is called a *subgradient*. Geometrically, subgradients correspond to supporting hyperplanes of the epigraph of v at (θ^0, δ^0).

Theorem 2.37 (Value-function characterization of KKT multipliers)**.** *Assume Slater's condition holds for the inequality constraints. Then*

$$\partial v(\theta^0, \delta^0) = -\Big\{ (\mu, \nu) : D(\mu, \nu) = v(\theta^0, \delta^0) \Big\}. \tag{2.69}$$

Proof. By Definition 2.36, a vector $(s, u) \in \partial v(\theta^0, \delta^0)$ if and only if

$$v(\theta, \delta) \geq v(\theta^0, \delta^0) + \langle s, \theta - \theta^0 \rangle + \langle u, \delta - \delta^0 \rangle \quad \text{for all } (\theta, \delta).$$

For convex functions, this condition is equivalent to the Fenchel–Young equality

$$v(\theta^0, \delta^0) + v^*(s, u) = \langle s, \theta^0 \rangle + \langle u, \delta^0 \rangle.$$

Setting $(s, u) = (-\mu, -\nu)$ and invoking Lemma 2.35, the above equality holds if and only if

$$D(\mu, \nu) = v(\theta^0, \delta^0).$$

Under Slater's condition, strong duality guarantees the existence of such multipliers, completing the proof. □

Corollary 2.38 (Alignment with complete Lagrangian optimality)**.** *Assume Slater's condition, LICQ at $x^\star$, and differentiability of all functions. Let $(\mu^\star, \nu^\star)$ be the KKT multipliers. Then the value*

function v is differentiable at $(0,0)$ and

$$\nabla v(0,0) = (-\mu^\star, -\nu^\star).$$

Equivalently, $(x^\star, \mu^\star, \nu^\star)$ is a saddle point of the complete Lagrangian $\mathcal{L}$ over

$$\mathbb{R}^n \times \mathbb{R}_+^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}.$$

3 Introduction to Non-Smooth Convex Programs

Optimality of a feasible point in a convex program can be characterized using saddle-point conditions regardless of whether the objective and constraint functions are differentiable. An alternative and more general approach relies on the notions of *subgradients* and *subdifferentials*, which extend the concept of gradients to nonsmooth convex functions.

3.1 Subgradients and Subdifferentials

Definition 3.1 (Subgradient)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a function and let $x^\star \in \mathbb{R}^n$. A vector $h \in \mathbb{R}^n$ is called a *subgradient* of f at $x^\star$ if

$$f(x) - f(x^\star) \geq h^\top (x - x^\star) \quad \forall x \in \mathbb{R}^n. \tag{3.1}$$

Definition 3.2 (Subdifferential)**.** The set of all subgradients of f at $x^\star$ is called the *subdifferential* of f at $x^\star$ and is denoted by

$$\partial f(x^\star).$$

Geometrically, $\partial f(x^\star)$ consists of the slopes of all supporting hyperplanes to the graph of f at $x^\star$.

If f is convex and differentiable at $x^\star$, then the subdifferential reduces to a singleton:

$$\partial f(x^\star) = \{\nabla f(x^\star)\}.$$

Thus, the concept of subgradients strictly generalizes classical gradients.

Example 3.3 (Absolute Value Function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = |x|.$$

The function f is convex but not differentiable at $x^\star = 0$. For $x^\star = 0$, a vector $h \in \mathbb{R}$ is a subgradient if and only if

$$|x| - |0| \geq h(x - 0) \quad \forall x \in \mathbb{R}.$$

This inequality holds precisely when $h \in [-1, 1]$. Hence,

$$\partial f(0) = [-1, 1].$$

Geometrically, each $h \in [-1, 1]$ corresponds to the slope of a supporting line to the graph of $|x|$ at the origin. For $x^\star \neq 0$, the function is differentiable and

$$\partial f(x^\star) = \{\text{sign}(x^\star)\}.$$

Example 3.4 (Maximum of Affine Functions)**.** Let

$$f(x) := \max\{a_1^\top x + b_1,\ a_2^\top x + b_2\},$$

where $a_1, a_2 \in \mathbb{R}^n$ and $b_1, b_2 \in \mathbb{R}$.

The function f is convex but generally nondifferentiable at points where both affine functions are active. If

$$a_1^\top x^\star + b_1 = a_2^\top x^\star + b_2,$$

then the subdifferential is

$$\partial f(x^\star) = \text{conv}\{a_1, a_2\},$$

the convex hull of the active gradients.

If only one affine function is active at $x^\star$, say $f(x^\star) = a_1^\top x^\star + b_1$, then

$$\partial f(x^\star) = \{a_1\}.$$

Thus, the subdifferential aggregates the gradients of all active components.

Example 3.5 (Indicator Function of a Convex Set)**.** Let $C \subset \mathbb{R}^n$ be a closed convex set and define the indicator function

$$\delta_C(x) := \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C. \end{cases}$$

The function δ_C is convex. Fix a point $x^\star \in C$. By definition, a vector $v \in \mathbb{R}^n$ belongs to the subdifferential $\partial\delta_C(x^\star)$ if and only if

$$\delta_C(x) - \delta_C(x^\star) \geq v^\top(x - x^\star) \quad \forall x \in \mathbb{R}^n. \tag{3.2}$$

Since $\delta_C(x^\star) = 0$, we distinguish two cases. If $x \notin C$, then $\delta_C(x) = +\infty$ and (3.2) holds automatically. If $x \in C$, then $\delta_C(x) = 0$ and (3.2) reduces to

$$0 \geq v^\top(x - x^\star) \quad \forall x \in C.$$

Therefore,

$$\partial\delta_C(x^\star) = \left\{ v \in \mathbb{R}^n \;\middle|\; v^\top(x - x^\star) \leq 0 \; \forall x \in C \right\},$$

which is precisely the normal cone to C at $x^\star$:

$$\partial\delta_C(x^\star) = N_C(x^\star).$$

This example shows that normal cones arise naturally as subdifferentials of indicator functions of convex sets, providing a direct link between subgradient theory and geometric optimality conditions.

Example 3.6 (Negative Entropy)**.** Let $f : \mathbb{R}^n_+ \to \mathbb{R}$ be the negative entropy

$$f(p) = \sum_{i=1}^{n} p_i \log p_i.$$

On the interior of the positive orthant, f is differentiable and

$$\partial f(p) = \{\nabla f(p)\}, \qquad \nabla f(p)_i = 1 + \log p_i.$$

At boundary points where $p_i = 0$ for some i, the gradient does not exist, but the subdifferential remains well-defined (possibly unbounded), reflecting the infinite slope of $x \log x$ near zero. This example illustrates how subgradients extend entropy-based optimization to boundary points of the simplex.

Theorem 3.7 (Existence of Subgradients)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is convex on a convex set $C \subset \mathbb{R}^n$, then the subdifferential $\partial f(x^\star)$ is nonempty for every $x^\star \in \operatorname{int}(C)$. Moreover, $\partial f(x^\star)$ is a compact convex set.*

Proof. Fix a point $x^\star \in \mathrm{int}(C)$. Consider the epigraph of f,

$$\mathrm{epi}(f) := \{(x,t) \in \mathbb{R}^{n+1} \mid t \geq f(x)\},$$

which is a closed convex subset of $\mathbb{R}^{n+1}$ since f is convex.

Because $x^\star \in \mathrm{int}(C)$, the point

$$(x^\star, f(x^\star))$$

lies on the boundary of $\mathrm{epi}(f)$, and there exist points strictly below the epigraph in a neighborhood of $x^\star$. By the supporting hyperplane theorem, there exists a nonzero vector $(h,-1) \in \mathbb{R}^{n+1}$ such that

$$(h,-1)^\top \big((x,t) - (x^\star, f(x^\star))\big) \leq 0 \quad \forall (x,t) \in \mathrm{epi}(f).$$

Expanding the inequality yields

$$h^\top (x - x^\star) - (t - f(x^\star)) \leq 0 \quad \forall t \geq f(x),$$

which implies

$$f(x) - f(x^\star) \geq h^\top (x - x^\star) \quad \forall x \in \mathbb{R}^n.$$

Thus, $h \in \partial f(x^\star)$, showing that the subdifferential is nonempty.

Next, we show that $\partial f(x^\star)$ is convex. Let $h_1, h_2 \in \partial f(x^\star)$ and let $\lambda \in [0,1]$. Then for all x,

$$f(x) - f(x^\star) \geq h_1^\top (x - x^\star), \qquad f(x) - f(x^\star) \geq h_2^\top (x - x^\star).$$

Taking a convex combination gives

$$f(x) - f(x^\star) \geq \big(\lambda h_1 + (1-\lambda) h_2\big)^\top (x - x^\star),$$

so $\lambda h_1 + (1-\lambda) h_2 \in \partial f(x^\star)$. Hence, $\partial f(x^\star)$ is convex.

Finally, we show compactness. Since $x^\star \in \mathrm{int}(C)$, there exists $\varepsilon > 0$ such that $B_\varepsilon(x^\star) \subset C$. For any $h \in \partial f(x^\star)$ and any u with $\|u\| = 1$, applying the subgradient inequality at $x^\star + \varepsilon u$ gives

$$f(x^\star + \varepsilon u) - f(x^\star) \geq \varepsilon h^\top u.$$

Thus,

$$h^\top u \leq \frac{f(x^\star + \varepsilon u) - f(x^\star)}{\varepsilon}.$$

Since the right-hand side is finite and uniformly bounded over the unit sphere, the set $\partial f(x^\star)$ is bounded.

Because $\partial f(x^\star)$ is closed (as an intersection of closed half-spaces) and bounded, it is compact.

This completes the proof. □

Theorem 3.8 (Basic Subdifferential Calculus)**.** *Let $f, f_1, f_2 : \mathbb{R}^n \to \mathbb{R}$ be convex functions, and let $\alpha \geq 0$ be a scalar. Then, for every $x^\star \in \mathbb{R}^n$:*

1. ***Positive scaling:***

$$\partial(\alpha f)(x^\star) = \alpha\, \partial f(x^\star).$$

2. ***Additivity:***

$$\partial(f_1 + f_2)(x^\star) = \partial f_1(x^\star) + \partial f_2(x^\star),$$

where the sum on the right-hand side is the Minkowski sum of sets.

3.2 Non-Smooth Convex Optimality

Theorem 3.9 (Equivalent First-Order Optimality Conditions)**.** *Let $C \subset \mathbb{R}^n$ be a convex set and let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. For a point $x^\star \in C$, the following statements are equivalent:*

(i) $x^\star$ is an optimal solution of

$$\min_{x \in C} f(x).$$

(ii) There exists a subgradient $h \in \partial f(x^\star)$ such that

$$h^\top (x - x^\star) \geq 0, \qquad \forall x \in C. \tag{3.3}$$

(iii) There exists $h \in \partial f(x^\star)$ such that

$$-h \in N_C(x^\star). \tag{3.4}$$

(iv) The following set inclusion holds:

$$0 \in \partial f(x^\star) + N_C(x^\star). \tag{3.5}$$

Proof. $(i) \Rightarrow (ii)$ If $x^\star$ is optimal, then $f(x) \geq f(x^\star)$ for all $x \in C$. By convexity, there exists $h \in \partial f(x^\star)$ such that

$$f(x) - f(x^\star) \geq h^\top (x - x^\star), \quad \forall x \in \mathbb{R}^n.$$

Restricting to $x \in C$ yields (3.3).

$(ii) \Leftrightarrow (iii)$ By definition of the normal cone,

$$-h \in N_C(x^\star) \quad \Longleftrightarrow \quad h^\top (x - x^\star) \geq 0, \ \forall x \in C.$$

$(iii) \Leftrightarrow (iv)$ Condition (3.4) holds if and only if there exists $h \in \partial f(x^\star)$ and $w \in N_C(x^\star)$ such that $h + w = 0$, which is exactly (3.5).

$(ii) \Rightarrow (i)$ If (3.3) holds, then using the subgradient inequality,

$$f(x) \geq f(x^\star) + h^\top (x - x^\star) \geq f(x^\star), \quad \forall x \in C,$$

so $x^\star$ is optimal. □

Theorem 3.10 (First-Order Optimality without Slater, Nonsmooth Case)**.** *Consider the convex optimization problem (2.60). Assume that all functions are convex (not necessarily differentiable), and let $\mathcal{F}^=$ denote the base feasible set. Let $x^\star \in \mathcal{F}^=$ be feasible.*

Then $x^\star$ is an optimal solution if and only if there exists a multiplier vector

$$\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}^<|}$$

and subgradients

$$h \in \partial f(x^\star), \qquad h_j \in \partial g_j(x^\star), \ j \in \mathcal{Q}^<,$$

such that

$$h + \sum_{j \in \mathcal{Q}^<} \mu_j^\star h_j \ \in \ N_{\mathcal{F}^=}(x^\star). \tag{3.6}$$

Proof. By Theorem 2.1, $x^\star$ is optimal if and only if there exists $\mu^\star \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that $x^\star$ minimizes the restricted Lagrangian

$$\mathcal{L}^c(x, \mu^\star) = f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j^\star g_j(x)$$

over $\mathcal{F}^=$.

Since $\mathcal{L}^c(\cdot, \mu^\star)$ is convex and $\mathcal{F}^=$ is convex, Lemma **??** applies. Thus, $x^\star$ minimizes $\mathcal{L}^c(\cdot, \mu^\star)$ over $\mathcal{F}^=$ if and only if there exists

$$g \in \partial_x \mathcal{L}^c(x^\star, \mu^\star)$$

such that

$$g^\top (x - x^\star) \geq 0, \qquad \forall x \in \mathcal{F}^=.$$

The subdifferential sum rule for convex functions yields

$$\partial_x \mathcal{L}^c(x^\star, \mu^\star) = \partial f(x^\star) + \sum_{j \in \mathcal{Q}^<} \mu_j^\star \partial g_j(x^\star).$$

Hence, there exist $h \in \partial f(x^\star)$ and $h_j \in \partial g_j(x^\star)$ such that

$$g = h + \sum_{j \in \mathcal{Q}^<} \mu_j^\star h_j.$$

The inequality

$$g^\top (x - x^\star) \geq 0 \quad \forall x \in \mathcal{F}^=$$

is equivalent to

$$g \in N_{\mathcal{F}^=}(x^\star),$$

which proves (3.6). □

Remark 3.11. If all functions are differentiable at $x^\star$, then each subdifferential reduces to a singleton containing the corresponding gradient. In this case, the condition (3.6) reduces to the normal-cone optimality condition stated in Theorem 2.4.

No constraint qualification, including Slater's condition, is required for this result. Inequality constraints that act as equalities on the feasible set are absorbed into the base set $\mathcal{F}^=$, while inactive inequality constraints do not contribute to the stationarity condition.

The normal-cone formulation emphasizes the geometric nature of optimality. Stationarity means that there exists no feasible descent direction at $x^\star$, or equivalently, that the negative subgradient of the objective lies in the normal cone to $\mathcal{F}^=$, which is the polar of the tangent cone $T_{\mathcal{F}^=}(x^\star)$.

Corollary 3.12 (Non-Smooth Karush–Kuhn–Tucker Conditions)**.** *Consider the convex optimization problem (2.60). Assume that all functions are convex (not necessarily differentiable) and that the inequality constraints satisfy Slater's condition. Equivalently, assume that the base feasible set satisfies*

$\mathcal{F}^{=} = \mathbb{R}^n$.

Then, a feasible point $x^\star$ is an optimal solution if and only if there exist multipliers

$$\mu_j^\star \geq 0, \quad j \in \mathcal{Q},$$

such that

$$0 \in \partial f(x^\star) + \sum_{j \in \mathcal{Q}} \mu_j^\star \, \partial g_j(x^\star). \tag{3.7}$$

Example 3.13 (ℓ^1–ℓ^∞ Norms and Non-Smooth KKT Conditions)**.** Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \|x\|_1 := \sum_{i=1}^n |x_i| \\ \text{s.t.} \quad & \|x\|_\infty := \max_{1 \leq i \leq n} |x_i| \leq 1. \end{aligned} \tag{3.8}$$

The feasible set is the ℓ^∞ unit ball $C := \{x \in \mathbb{R}^n \mid \|x\|_\infty \leq 1\}$, which is compact, convex, and has nonempty interior. Hence Slater's condition holds. Since $\|x\|_1 \geq 0$ for all x and $\|0\|_1 = 0$, the unique optimal solution is $x^\star = 0$. The subdifferential of the ℓ^1 norm at the origin is given by

$$\partial \|x\|_1(0) = \{v \in \mathbb{R}^n \mid \|v\|_\infty \leq 1\}.$$

The inequality constraint can be written as

$$g(x) := \|x\|_\infty - 1 \leq 0,$$

whose subdifferential at $x^\star = 0$ is

$$\partial g(0) = \partial \|x\|_\infty(0) = \operatorname{conv}\{\pm e_1, \ldots, \pm e_n\},$$

where $\{e_i\}$ denotes the standard basis of $\mathbb{R}^n$.

Since $x^\star = 0$ lies in the interior of the feasible set C, the normal cone satisfies

$$N_C(0) = \{0\}.$$

By the non-smooth Karush–Kuhn–Tucker conditions, optimality requires the existence of a multiplier $\mu^\star \geq 0$ such that

$$0 \in \partial \|x\|_1(0) + \mu^\star \partial g(0).$$

Choosing $\mu^\star = 0$ yields

$$0 \in \partial \|x\|_1(0),$$

which holds. Therefore, $x^\star = 0$ satisfies the non-smooth KKT conditions and is the optimal solution of (3.8). This example illustrates the dual geometric relationship between the ℓ^1 and ℓ^∞ norms: the subdifferential of $\|\cdot\|_1$ is an ℓ^∞ ball, while the subdifferential of $\|\cdot\|_\infty$ is an ℓ^1 polytope.

4 Further Reading

A comprehensive and unifying framework for convex analysis, including faithfully convex functions and duality, is developed in [10]. Cone-based formulations of feasibility, consistency, and optimality conditions are presented in [5, 4]. Theorems of the alternative and their role in optimality and duality are treated in [9]. Modern treatments of Lagrangian duality, saddle-point theory, and constraint qualifications are given in [6, 7]. A general variational framework for optimality conditions beyond standard constraint qualifications is presented in [8]. Subdifferential calculus and non-smooth convex programming are treated in [1, 2, 3, 8], together with numerical methods such as penalty, barrier, SUMT, and augmented Lagrangian methods; see also [6].

5 Exercise

Exercise 1 (Locally Flat Surfaces for Faithfully Convex Functions)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable *faithfully convex* function of the form

$$f(x) = \varphi(Ax + b) + a^T x + \alpha,$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $a \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, and $\varphi : \mathbb{R}^m \to \mathbb{R}$ is strictly convex.

Recall that f is said to have a *locally flat surface* (LFS) at $x^\star \in \mathbb{R}^n$ if

$$\mathcal{N}(\nabla f(x^\star)) = \mathcal{D}^= f(x^\star),$$

where $\mathcal{D}^= f(x^\star)$ denotes the cone of directions along which f is locally constant at $x^\star$.

(i) Show that

$$\nabla f(x) = A^T \nabla \varphi(Ax + b) + a.$$

(ii) Prove that a direction $d \in \mathbb{R}^n$ satisfies $f(x^\star + td) = f(x^\star)$ for all sufficiently small t if and only if

$$Ad = 0 \quad \text{and} \quad a^T d = 0.$$

(iii) Show that if $\nabla f(x^\star) = 0$, then f has the LFS property at $x^\star$ if and only if f is constant.

(iv) Suppose $\nabla f(x^\star) \neq 0$. Prove that f has the LFS property at $x^\star$ if and only if

$$\operatorname{rank} \begin{bmatrix} A \\ a^T \end{bmatrix} = 1.$$

(v) Briefly interpret the rank condition above in terms of the geometry of the level sets of f near $x^\star$.

Exercise 2 (Parametric Linear Program and Optimal Value Function)**.** Consider the linear programming problem parameterized by $\theta \in \mathbb{R}$ and $\delta \in \mathbb{R}$:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \langle c, x \rangle \\ \text{s.t.} \quad & \langle a, x \rangle \geq \theta, \\ & 0 \leq x_j \leq \delta, \quad j = 1, \ldots, n, \end{aligned} \tag{5.1}$$

where the vectors $c \in \mathbb{R}^n$ and $a \in \mathbb{R}^n$ are fixed, and the box constraint $0 \leq x \leq \delta$ is understood componentwise. Assume throughout that $\delta \geq 0$.

(i) Determine for which values of (θ, δ) the feasible set of (5.1) is nonempty. In particular, compute the maximum value of $\langle a, x \rangle$ over the box $0 \leq x \leq \delta$, and characterize the largest admissible θ as a function of δ.

(ii) Show that if the problem is feasible, then an optimal solution can be chosen at an extreme point of the feasible polyhedron. Describe the structure of these extreme points.

(iii) Assume the problem is feasible. Show that an optimal solution is obtained by setting

$$x_j = \begin{cases} 0, & c_j > 0, \\ \delta, & c_j < 0, \end{cases}$$

except possibly for a subset of indices that must be adjusted in order to satisfy the constraint $\langle a, x \rangle \geq \theta$.

(iv) Define the optimal value function

$$v(\theta, \delta) = \inf \left\{ \langle c, x \rangle : \langle a, x \rangle \geq \theta,\ 0 \leq x_j \leq \delta,\ j = 1, \ldots, n \right\}.$$

Show that $v(\theta, \delta)$ is a convex, piecewise linear function of (θ, δ).

(v) Assume the problem is feasible and bounded. Interpret the partial derivatives (or subgradients) of $v(\theta, \delta)$ with respect to θ and δ in terms of the Lagrange multipliers associated with the constraints $\langle a, x \rangle \geq \theta$ and $x_j \leq \delta$.

(vi) Analyze explicitly the case $a = \mathbf{1}$ (the all-ones vector). Give an explicit formula for $v(\theta, \delta)$ and describe how the set of active constraints depends on θ.

Exercise 3 (Failure of Slater's Condition in a Degenerate Convex Program)**.** Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_2 \\ \text{s.t.} \quad & x_2 \geq x_1^2, \\ & x_2 \leq 0. \end{aligned} \tag{5.2}$$

(i) Show that the feasible set of (5.2) is nonempty and convex, and identify it explicitly.

(ii) Show that Slater's condition does not hold for this problem. In particular, show that there is no point satisfying both inequality constraints strictly.

(iii) Solve (5.2) and determine the unique optimal solution and optimal value.

(iv) Describe the geometry of the feasible set and explain why it has empty interior relative to $\mathbb{R}^2$. Relate this to the notion of constraint degeneracy.

(v) Form the Lagrangian and derive the KKT conditions. Determine whether Lagrange multipliers exist that satisfy the KKT system.

(vi) Derive the Lagrange dual problem explicitly. Show that strong duality holds despite the failure of Slater's condition.

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